
MUTSOTSO WANYU WIRI NA TIMA
YOUR STOVE HAS VALUE
O SEU FOGÃO TEM VALOR
Mutsotso wanyu omalisa nthena epacherrwa anyu
How o seu fogäie creates money yocẽ
Elapho muraka wa apithaaka mukhaya A guit do home A guide you can keep do home

SECTION 1: EXECUTIVE SUMMARY

CARBON CLARITY

Fair, Transparent Carbon Markets for Mozambique Communities

Executive Summary

The Problem We Discovered

In 2024, IOM Mozambique distributed improved cookstoves to displaced communities across five pilot sites. The project worked well, households used the stoves, emissions dropped, and the technical team collected all the right data. But the project never registered for carbon credits. Why? The carbon developer had "integrity issues." The communities who used those stoves every day never saw a single cent from carbon finance.

This is not an isolated incident. Across Africa, fewer than 30% of carbon projects publish clear benefit-sharing formulas. Even when projects do succeed and credits are sold, communities rarely understand what price their credits fetched, what costs were deducted, or what margin the developers kept. They are told to trust the system. But when the system is designed to keep them in the dark, that trust breaks.

Here is the reality: a displaced woman in Cabo Delgado cooks with an improved stove that generates carbon credits worth $24. She might receive $3, or she might receive nothing at all. She has no receipt, no breakdown, no way to verify if the number is fair. And when carbon projects fail because of developer misconduct, she is the one who loses.

This is the gap we are closing.

Our Solution: Transparency as Infrastructure

Carbon Clarity is not another carbon project. We are a transparency and education layer that works inside existing carbon frameworks, Gold Standard, Verra, IOM partnerships, to ensure communities understand the carbon process and can verify they are treated fairly.

We do two things:

First, we educate. Through our Carbon Clarity Hub, we use VSLA meetings, visual guides, community radio, and public price boards to teach communities what carbon credits are, how the process works, and what fair benefit-sharing looks like. We do not wait until Year 3 when credits are sold. We start on Day 1, building carbon literacy while stoves are being distributed.

Second, we create accountability. When carbon credits are eventually sold, we provide the tools communities need to hold developers accountable: itemized receipts showing price per credit, costs, and margins; public price boards updated every quarter; and VSLA-linked oversight committees that can question numbers and raise grievances.

We do not replace the carbon developer. We do not handle registration, monitoring, or sales. We make sure the developer's promises to communities are transparent, understandable, and enforceable.

Why This Matters Now

IOM is scaling up. They are distributing 2,770 new cookstoves across nine additional sites in Mozambique. This time, they want to get carbon finance right. They want communities to benefit fairly. And they want to avoid another failed pilot.

That is where Carbon Clarity comes in. We provide the transparency infrastructure that prevents integrity failures, builds community trust, and ensures that when carbon revenue finally flows, two or three years from now, communities are not signing blind. They know what to expect, they can verify what they receive, and they have clear channels to challenge anything that feels wrong.

Beyond Mozambique, this model matters for Africa's carbon future. The continent could generate $6 billion annually from carbon markets by 2030. But only if communities trust the system. Transparency is not a nice-to-have. It is the foundation that makes scale possible.

We deploy these tools in three phases over 36 months, starting simple and building complexity only as communities demonstrate understanding and request more detail.

What We Deliver

In partnership with IOM Mozambique and carbon project developers, Carbon Clarity will:

- Train 14 active VSLA groups in carbon literacy using visual, accessible materials
- Provide transparent reporting tools (price boards, receipts, benefit dashboards) aligned with Gold Standard requirements
- Build community oversight structures so elected committees can verify developer claims

- ❖ Establish clear grievance mechanisms with documented response times and public outcomes
- ❖ Pilot in IOM's scale-up sites (2,770 cookstoves, nine locations) starting in 2026

Expected impact:

- Communities understand carbon finance before credits are issued
- Fair benefit-sharing is verified, not just promised
- Developer integrity is enforced through public accountability
- IOM's scale-up avoids the failures of the pilot phase
- A replicable model emerges for transparent carbon finance in displacement settings across Africa

Why Carbon Clarity Will Win

We solve a problem that IOM has already experienced firsthand. We work within established standards, not against them. We have a confirmed partnership path with IOM Mozambique. We have 14 existing VSLA groups ready to pilot. And we have designed for the real constraints of displacement contexts, low literacy, limited connectivity, high mobility, and fragile trust.

This is not a concept. This is a blueprint ready to test with communities who are already waiting for their cookstoves.

The question is not whether transparency is needed. IOM's failed pilot proves it is. The question is whether we build it now, before the next 2,770 households are left in the dark.

SECTION 2: PROBLEM STATEMENT

When Carbon Projects Fail, Communities Pay the Price

A Failure That Should Not Have Happened

In 2024, IOM Mozambique worked with a carbon project developer to distribute improved cookstoves across five pilot sites in northern Mozambique. The stoves worked. Families used them every day. Field staff monitored usage rates and fuel consumption. The data showed clear emissions reductions, exactly what carbon markets reward.

But the project never registered for carbon credits. The developer had "integrity issues," and the registration never went through. The communities who cooked with those stoves every single day, who did everything they were supposed to do, never received a single meticais from carbon finance.

This is not a story about technology failure or bad weather or community resistance. This is a story about a system that was supposed to benefit communities but left them with nothing. And the hardest part? The communities did not even understand what went wrong. They just knew that the "carbon money" they were told to expect never came.

When we interviewed IOM Mozambique about this experience, they were direct: *"There were integrity issues with the carbon developer, so the project did not end up being registered."* Five sites. Hundreds of households. Zero carbon revenue. And no clear explanation communities could understand.

This failure is our starting point. Because if it happened once in Mozambique, it can happen again. And it is happening, right now, in carbon projects across the continent.

The Pattern is Bigger Than One Failed Pilot

IOM's experience in Mozambique is not unique. It is a symptom of a much larger problem in how carbon markets treat the communities who create the value.

Here is what we know from research and practice:

Fewer than 30% of carbon projects globally publish clear benefit-sharing formulas. That means in 70% of projects, communities have no idea what they should expect to receive, what costs are legitimate, or what margins intermediaries are taking. They are told "trust us" and handed a long contract in a language many cannot read.

Even when projects do succeed technically—when credits are issued and sold—the money rarely flows transparently. In our research on benefit-sharing arrangements in voluntary carbon markets, we found that most projects provide no breakdown to communities showing: the price per credit, the costs that were deducted, the developer's margin, or how the final payment amount was calculated. Payments simply appear—or do not appear—and communities have no way to verify if the number is fair.

IOM Mozambique confirmed this reality in our interview. Mercy, who works directly with these communities, told us: *"Even in fully registered cookstove carbon projects globally, it is not common practice for developers to provide communities with a breakdown of credit pricing, deductions, or revenue flows. The industry acknowledges a lack of transparency in benefit-sharing."*

Let that sink in. The industry acknowledges the problem. And yet it continues.

What This Looks Like on the Ground

To understand what this means in practice, consider a typical scenario:

Amina is a 24-year-old displaced woman living in a resettlement site in Cabo Delgado. She uses an improved cookstove every day. A field officer visits occasionally and asks questions about her usage. One day, maybe two years later, 500 meticais appears on her mobile money account. No receipt. No explanation. No context.

Amina has no idea:

- How many carbon credits her stove generated
- What price those credits sold for
- What costs were deducted (transport? verification? developer margin?)
- Whether 500 meticais is 10% of the value or 90%
- Whether her neighbor received more or less, and why
- Who to ask if she thinks the number is wrong

She is told this is "benefit-sharing." But from her perspective, it looks more like charity. She has no agency, no information, no power to negotiate. She either accepts what she receives, or she walks away from the project entirely.

This is not hypothetical. This is how most cookstove carbon projects operate today.

Why Transparency Matters More Than We Think

Some might ask: if communities are receiving some money, why does transparency matter? Why not just trust that developers are doing their best?

Here are three reasons why transparency is not optional—it is foundational:

First, trust. Without transparency, communities have no reason to trust carbon projects. And without trust, participation drops. In our interview, IOM flagged "low trust or poor communication" as one of the main reasons carbon and livelihood projects fail in displacement settings. When communities do not understand what is happening or feel excluded, "engagement drops and misinformation spreads." Transparency is how you build and maintain trust over the long term.

Second, fairness. Transparency is the only way communities can verify they are being treated fairly. Right now, developers can say "we gave you 30% of revenue" and communities have no way to check if that is true. Was it 30% of gross revenue or net revenue? What costs were deducted first? Was the margin capped or unlimited? Without transparent numbers, "fair" is just a word with no substance.

Third, scale. Africa is being told it can generate $6 billion annually from carbon markets by 2030. But that depends on communities choosing to participate. If the first generation of carbon projects leaves communities feeling exploited or confused, they will not sign up for the second generation. Transparency is not just about fixing past wrongs—it is about making future scale possible.

The Specific Barriers in Displacement Settings

The transparency problem is even harder in displacement contexts like northern Mozambique. Here is why:

Literacy is limited. In Cabo Delgado, female literacy is estimated at 30-35%. Most people in these areas speak local languages like Emakhuwa, not Portuguese. Yet carbon project contracts are often long, technical documents in Portuguese or English. Communities are signing agreements they cannot read, let alone understand. Mercy confirmed this in our interview: *"Majority of the people in these areas speak local languages, not Portuguese. So documents are translated to local languages, and the developer with IOM oversight always explains complex topics."* But translation and explanation are not enough if the underlying information—prices, costs, margins, is never shared in the first place.

Power dynamics are unequal. Displaced communities are already in a vulnerable position. They depend on humanitarian organizations for services, and they may fear that asking too many questions will result in exclusion from programs. This creates a dynamic where "consent" becomes more about going along to get along, rather than making an informed choice. Clara Maria, one of the judges who reviewed our work, put it clearly: *"In this displacement context where trust and power dynamics and literacy levels may vary, how would you ensure that women, youth, and the most vulnerable households can actually participate meaningfully in the*

community structures?" This is the challenge: transparency is not just about posting numbers. It is about creating conditions where people feel safe to question those numbers.

There is no accountability infrastructure. In many carbon projects, there is no clear process for raising complaints or challenging payment amounts. Communities do not know who to go to if something feels wrong. And even when complaints are raised, there is often no documented response or public outcome. The system is designed to be opaque, and that opacity protects developers more than it protects communities.

The Cost of Getting This Wrong

What happens when carbon projects fail to provide transparency?

We already know from IOM's pilot: the project does not register, or it collapses, and communities are left with nothing. But even when projects technically succeed, lack of transparency creates long-term harm:

- Communities lose income they are entitled to. If a developer says they paid "fair market price" but communities cannot verify it, they may be receiving 20% when they should be receiving 60%.
- Trust erodes. Once burned, communities become skeptical of all future carbon projects. This makes it harder for good actors to operate, because they inherit the mistrust created by bad actors.
- Vulnerable groups are excluded. When transparency is missing, the people with the least power, women, youth, people with disabilities, are the first to be sidelined. Elite capture becomes easier when no one can see the numbers.
- Africa's carbon potential stalls. If the first wave of projects leaves communities feeling exploited, the second wave will struggle to find participants. Transparency today is what makes scale possible tomorrow.

The Gap We Are Addressing

So here is the core problem, stated clearly:

Communities in displacement settings are being asked to participate in carbon markets without the information, tools, or power to verify they are being treated fairly. Contracts are incomprehensible. Payments are unexplained. Grievance mechanisms are weak or nonexistent. And when projects fail due to developer misconduct, communities, who did everything right, are the ones who lose.

This is not a small gap. This is a foundational flaw in how carbon markets operate in vulnerable contexts.

Gold Standard and Verra have benefit-sharing requirements. But those requirements focus on what developers must promise, not what communities must be able to verify. There is a difference between a developer saying "we will share benefits fairly" and a community being able to check the math.

That difference, the gap between promise and proof—is what Carbon Clarity exists to close.

Why Now? Why Mozambique?

IOM Mozambique is about to scale up. They are distributing 2,770 new cookstoves across nine additional sites. This time, they want to avoid the mistakes of the pilot. They want communities to understand the carbon process from the beginning. They want benefit-sharing to be transparent, not just promised. And they want to work with partners who can provide the education and accountability infrastructure that prevents future failures.

This is our moment. We have a confirmed partner in IOM. We have 14 active VSLA groups ready to pilot transparency tools. We have a clear understanding of what went wrong in the pilot and how to fix it. And we have a 2-3 year window before carbon revenues flow, which is exactly when education and preparation matter most.

If we wait until credits are sold to think about transparency, it is too late. Transparency must be built from Day 1.

SECTION 3: TARGET AUDIENCE

Who We Serve and Why

Understanding Who Creates Carbon Value

Carbon Clarity is not designed for everyone. It is designed specifically for communities and individuals who are creating carbon value through their daily actions but lack the information and power to benefit fairly from that value.

When we started this work, we thought narrowly: cookstove users. But our research with IOM Mozambique showed us the opportunity is broader. There are at least two groups in northern Mozambique who are already generating, or could generate, carbon credits, but who face the same transparency gap.

Let us introduce them.

Primary Audience 1: Households Using Improved Cookstoves

Who They Are

These are displaced and host-community households living in resettlement sites across northern Mozambique, particularly in Cabo Delgado, Nampula, and Niassa provinces. The majority are women-headed households. Many are members of Village Savings and Loan Associations (VSLAs), which they use for small loans, savings, and community support.

IOM has already distributed improved cookstoves in pilot sites and is now scaling to 2,770 additional stoves across nine locations. These households use their stoves daily, not because of carbon finance, but because the stoves reduce the time and effort required to collect firewood and lower indoor smoke that harms children's lungs.

What They Generate

Each household using an improved cookstove can generate approximately 1.5 to 3 tonnes of CO_2 equivalent (tCO_2e) in verified emissions reductions per year. Over a typical 5-year project cycle, that is 7.5 to 15 tCO_2e per household. At current market prices of $10-15 per credit, that represents $75 to $225 in potential carbon value per household over five years.

That may not sound like much to an international buyer. But in a context where women earn $1-3 per day from informal work, $75 to $225 is meaningful income, if it reaches them fairly and transparently.

What They Need From Us

These households need three things:

First, education. They need to understand what carbon credits are, how the process works, and what they should expect. Right now, as Mercy told us, most households have no knowledge of carbon markets. They use stoves because stoves are useful, not because they understand the carbon finance that might follow years later. If we wait until Year 3 to educate them, we have already lost the opportunity to build informed consent.

Second, transparency. When carbon revenues eventually flow, they need to be able to verify the payments are fair. That means seeing the price per credit, understanding what costs were deducted, knowing what margin the developer kept, and having a clear breakdown of why they received the amount they did. Without this, "benefit-sharing" is just a promise with no proof.

Third, voice. They need structures that give them power to question, challenge, and renegotiate if something feels wrong. That means accessible grievance channels, community oversight committees, and public accountability mechanisms that make it safe to raise concerns without fear of exclusion.

Why VSLAs Matter

IOM confirmed that there are currently 14 active VSLA groups operating in the sites where the cookstove project is being scaled. This is critical for our model.

VSLAs are not just savings groups. They are trusted community institutions with strong internal accountability. VSLA members already track contributions and loans carefully, down to the meticais. They meet regularly, usually weekly or bi-weekly, which gives us a consistent channel for education and updates. And because VSLAs are member-owned and governed, they provide a structure we can build on rather than creating something new from scratch.

As Mercy put it: *"There is a strong potential for VSLAs to play a key role once we register a carbon project. They are trusted community institutions that can help share accurate information about carbon projects, support households to access financing for cookstoves and other energy solutions, and potentially serve as channels for distributing benefits or community-level incentives."*

This is why our entire model runs through VSLAs. They are not just a delivery channel—they are the governance backbone that makes transparency enforceable at the community level.

Primary Audience 2: Youth Producing Sustainable Charcoal and Briquettes

Who They Are

Youth in displacement-affected areas of northern Mozambique face unemployment rates above 60%. Most have completed some secondary education but have few formal job opportunities. Many are involved in informal activities, small trade, agriculture, casual labor—but struggle to earn stable income.

At the same time, deforestation in northern Mozambique is accelerating at rates of 0.8-1.2% annually, driven largely by demand for cooking fuel. Displaced communities and host populations are cutting trees to make charcoal, which creates a destructive cycle: more displacement leads to more demand for firewood, which leads to more forest loss, which worsens the environmental conditions that make livelihoods harder.

Some young people, especially young men, but increasingly young women, are starting to produce sustainable alternatives: briquettes made from agricultural waste (maize cobs, rice husks, sawdust) and improved charcoal produced through more efficient kilns that reduce emissions.

When we asked Mercy about this, she confirmed: *"Yes, they produce charcoal but they do not have knowledge of carbon markets."*

This is the opportunity. Youth are already creating carbon value through their production of sustainable cooking fuel. They just do not know it yet.

What They Generate

Sustainable charcoal and briquette production can generate carbon credits in two ways:

First, avoid deforestation. Every tonne of briquettes produced from agricultural waste is roughly one tonne of wood that does not need to be cut from forests. Depending on the carbon accounting methodology, this can generate 1 to 1.5 tCO_2e per tonne of briquettes.

Second, improved production methods. More efficient kilns and production processes reduce methane and black carbon emissions compared to traditional methods, which can generate additional credits.

A small youth-led briquette business producing 10 tonnes per month could generate 120 to 180 tCO_2e per year. At $10-15 per credit, that is $1,200 to $2,700 annually in potential carbon revenue, on top of income from selling the briquettes themselves.

What They Need From Us

Youth producers face the same transparency challenge as cookstove users, but with an added layer: they also need to understand that their work has carbon value in the first place.

First, awareness. Most youth do not know that sustainable charcoal or briquette production can generate carbon credits. Our education component needs to explain not just how carbon markets work, but why their specific activity qualifies and what that could mean for their income.

Second, transparency. When credits are sold, youth producers need the same verification tools as households: price boards, itemized receipts, and oversight structures that let them confirm they are receiving a fair share.

Third, market access and support. Unlike households who receive stoves for free, youth producers are running small businesses. They need startup support, production training, and connections to buyers for both their physical products (briquettes) and their carbon credits. Carbon Clarity can provide the transparency layer, but youth also need partners who can help with business development and market linkages.

Why Youth Matter for Scale

Youth are not just another target audience. They are a scaling mechanism.

If youth-led briquette production works in Mozambique, it solves two problems at once: unemployment and deforestation. It creates dignified, income-generating work for thousands of young people while reducing the pressure on forests that cookstove programs alone cannot address. And because youth are mobile, connected, and entrepreneurial, a successful model can spread faster through peer networks than through top-down programs.

This is why we include youth producers as a primary audience, not a secondary one. They are central to the long-term vision of what transparent carbon markets can enable in displacement settings.

Secondary Audience: Carbon Buyers and Development Partners

Who They Are

Our secondary audience includes:

- Carbon credit buyers (corporations, project developers, carbon funds) seeking high-integrity credits with documented community benefits
- Development partners (UN agencies, international NGOs, bilateral donors) funding climate and livelihood programs
- Government actors (Mozambican ministries of environment, energy, and social affairs) responsible for climate policy and displacement response

What They Need From Us

These actors are not the direct beneficiaries of Carbon Clarity, but they are essential enablers. They need to see that transparent, community-centered carbon projects are not just ethical—they are viable, scalable, and lower-risk than opaque alternatives.

For carbon buyers, transparency reduces reputational risk. In a market where NGOs and journalists are increasingly scrutinizing "greenwashing," buyers need proof that communities are genuinely benefiting and consenting. Our model provides that proof through public reporting, third-party verification, and community oversight. This makes credits more attractive and can justify premium pricing.

For development partners, transparency aligns with their mandates around protection, equity, and "do no harm." IOM cannot work with carbon developers who exploit communities, because that contradicts humanitarian principles. Carbon Clarity gives IOM and other partners a way to engage in carbon finance without compromising their values or their relationships with communities.

For the government, transparency supports national climate goals and benefit-sharing policies. As Mozambique develops its approach to carbon markets, transparent models like ours can provide a blueprint for how to ensure communities benefit fairly while still attracting investment.

Who We Do Not Serve

It is also important to be clear about who Carbon Clarity is not designed for:

We are not designed for developers who want to hide margins or avoid accountability. If a carbon project developer is not willing to publish prices, costs, and benefit-sharing formulas publicly, they should not be working in displacement settings. Our model makes that type of opacity

impossible, which means developers who rely on information asymmetry will not want to work with us. That is by design.

We are not designed for contexts where communities are already well-informed and empowered. If a community has high literacy, strong legal protections, and existing carbon market expertise, they may not need our tools. Carbon Clarity is specifically built for low-resource, fragile, displacement-affected contexts where information gaps and power imbalances are most severe.

We are not a replacement for MRV or carbon registration systems. We are a transparency layer, not a carbon project developer. If a partner is looking for someone to handle monitoring, verification, and credit issuance, that is not us. We make those processes transparent to communities, we do not do the processes ourselves.

Why These Audiences, Together, Create a System

The power of Carbon Clarity is not in serving one group in isolation. It is in creating a system where multiple actors, households, youth, VSLAs, developers, IOM, buyers, all benefit from transparency.

When households and youth can verify fair treatment, they stay engaged and participate consistently. When VSLAs have clear oversight roles, they become guardians of integrity. When developers operate transparently, they attract better buyers and reduce conflicts. When IOM can point to a transparent model, they can scale carbon finance with confidence. And when buyers see documented community benefits, they pay premium prices that increase the revenue available to share.

Transparency is not a zero-sum game. It creates value for everyone except those who depend on keeping communities in the dark.

That is who we serve: the actors who want carbon markets to work fairly. And that is who we exclude: the actors who profit from opacity.

SECTION 4: RESEARCH & METHODOLOGY

How We Learned What We Needed to Know

Starting With Questions, Not Solutions

When we began this work, we did not start with Carbon Clarity. We started with a hunch: that something was broken in how carbon markets engage communities in displacement settings. But hunches are not enough. We needed to understand the problem deeply before we could propose a solution that would actually work.

So we followed a design thinking process: Empathise → Define → Ideate → Prototype → Test. This approach forced us to listen before we spoke, to understand before we designed, and to validate our assumptions with real evidence rather than educated guesses.

Here is how that process unfolded.

Phase 1: Empathise - Understanding the Context

Secondary Research: Building the Foundation

We started with desk research to understand the landscape. We needed to answer basic questions: What do we know about carbon markets in Africa? What does the research say about benefit-sharing? What is the specific context of displacement in northern Mozambique?

Key sources we reviewed:

On carbon markets and integrity:

- Africa Carbon Markets Initiative (ACMI) reports on Africa's carbon potential and high-integrity requirements
- Integrity Council for the Voluntary Carbon Market (ICVCM) Core Carbon Principles
- Ecosystem Marketplace data on voluntary carbon market pricing and volumes

On benefit-sharing:

- "Assessing transparency and integrity of benefit-sharing arrangements in voluntary carbon markets" - a detailed study documenting opacity in existing projects
- Gold Standard's factsheet on inclusivity, safeguards and benefit sharing
- Forest Carbon Partnership Facility (FCPF) guidance on designing benefit-sharing arrangements

On displacement and livelihoods in Mozambique:

- IOM Mozambique displacement reports and energy project briefs
- World Bank data on energy access and literacy rates in Mozambique
- CARE and Oxfam evaluations of Village Savings and Loan Association (VSLA) programs

What we learned:

Three patterns emerged clearly from the literature:

First, the transparency gap is real and documented. Study after study confirmed that fewer than 30% of carbon projects publish clear benefit-sharing formulas, and communities rarely receive itemized breakdowns of pricing, costs, or margins. This is not anecdotal—it is a documented, industry-acknowledged problem.

Second, standards exist but lack enforcement mechanisms at the community level. Gold Standard and Verra require benefit-sharing plans, but they do not mandate that communities receive accessible, verifiable information. The gap is not in policy—it is in implementation and accountability.

Third, displacement contexts amplify the problem. Low literacy (30-35% for women in Cabo Delgado), language barriers (most speak Emakhuwa, not Portuguese), power imbalances, and fragile trust make informed consent nearly impossible without deliberate, accessible design. Standard approaches that might work in other contexts simply do not translate.

These insights shaped our problem statement, but they did not tell us what communities actually experience or what solutions might work in practice. For that, we needed primary research.

Primary Research: Listening to the Field

The most important research we conducted was a 90-minute structured interview with Mercy Mutavi from IOM Mozambique's livelihoods and energy team. Mercy works directly with the communities we aim to serve. She has been on the ground through the pilot project, the failure to register, and the planning for scale-up. Her insights were invaluable.

Why this interview mattered:

We could have designed Carbon Clarity based on desk research alone. But we would have gotten it wrong. The interview revealed critical details that changed our approach entirely.

What we asked:

We prepared 10 core questions organized around validation, feasibility, and partnership:

1. Are communities currently receiving carbon payments, and if so, do they get breakdowns?
2. Do households feel payments are fair, or do they complain about not understanding?
3. Are VSLAs active and could they realistically take on a carbon oversight role?
4. What percentage of households can read contracts, and what communication methods work?
5. Are youth producing briquettes or sustainable charcoal, and do they know about carbon?
6. How does carbon finance currently work—who handles credits, IOM or a developer?
7. Would IOM be open to piloting a transparent carbon model?
8. Which sites would be suitable for piloting?
9. What are the main reasons carbon projects fail in these areas?
10. What would be needed to move from concept to pilot?

What we learned, and how it changed our design:

Finding 1: The problem is worse than we thought.

Mercy confirmed: *"Communities are not receiving carbon payments because the cookstove project we implemented in 5 pilot sites did not get registered. There were integrity issues with the carbon developer so the project did not end up being registered."*

This was not hypothetical. IOM had experienced exactly the failure we were concerned about. And worse, Mercy added: *"Even in fully registered cookstove carbon projects globally, it is not common practice for developers to provide communities with a breakdown of credit pricing, deductions, or revenue flows."*

What this meant for our design: We were not solving an edge case. We were solving a systemic problem that IOM had lived through. This became the opening of our problem statement.

Finding 2: We had misunderstood our role.

Initially, we positioned Carbon Clarity as a carbon intermediary—we would buy credits from communities and sell them to buyers. But Mercy corrected us: *"Carbon pricing mechanisms are already defined by the major carbon standards such as Gold Standard and Verra, and any solution must align with those frameworks rather than create parallel systems."*

She explained that IOM does not handle carbon credits. The project developer is responsible for registration, MRV, credit issuance, and sales. IOM plays a facilitating and safeguarding role.

What this meant for our design: We are not a replacement intermediary. We are a transparency and education layer that works within the existing system. This shift was fundamental—it

changed our entire value proposition from "we buy and sell credits" to "we make the developer's process transparent to communities."

Finding 3: VSLAs are even more important than we realized.

Mercy confirmed: *"Yes, we currently have 14 active VSLA groups in each of the sites where our project is being implemented. There is a strong potential for VSLAs to play a key role once we register a carbon project."*

But she went further, explaining that VSLAs are not just channels—they are *"trusted community institutions that can help share accurate information, support households to access financing, and potentially serve as channels for distributing benefits."*

What this meant for our design: VSLAs became the backbone of our entire governance and education model, not just one delivery channel among many. Every tool we designed—training, liaisons, oversight committees, now runs through VSLAs.

Finding 4: Youth are producing charcoal but have no carbon knowledge.

When we asked about youth and briquettes, Mercy confirmed: *"Yes they produce charcoal but they do not have knowledge of carbon markets."*

This validated our hunch that youth employment and deforestation reduction could be addressed simultaneously through transparent carbon finance, but only if we educate youth first about the value their work creates.

What this meant for our design: We added youth as a full primary audience, not just a footnote. Our education tools now explicitly target young producers alongside cookstove users.

Finding 5: The timeline is longer than we thought.

Mercy was clear: *"Carbon revenues only flow after a project is fully registered, credits are issued, and sales occur, which can take 2-3 years depending on the standard."*

What this meant for our design: We cannot wait until Year 3 to engage communities. We need a phased education approach that starts on Day 1, builds literacy over time, and prepares communities to verify fairness when money eventually flows. This led directly to our three-phase rollout model.

Finding 6: The risks are specific and addressable.

When we asked what makes projects fail, Mercy listed: population mobility, security constraints, low trust, weak MRV, economic barriers, and elite capture.

What this meant for our design: We explicitly designed around each risk. VSLA-based models move with communities (addressing mobility). Public boards and receipts prevent elite capture. Multi-channel education (radio, visual guides, oral) addresses low trust and literacy barriers.

Finding 7: Partnership is confirmed.

When we asked if IOM would pilot our solution, Mercy said: *"Yes, the best idea from the hackathon will be refined together with IOM and our implementing partners and piloted during the project scale-up with a small group of communities."*

What this meant for our design: We have a real path to implementation. This is not theoretical. IOM is scaling to 2,770 cookstoves across nine sites, and they want transparency infrastructure. We have a partner, a timeline, and a mandate.

Phase 2: Define - Synthesizing the Problem

After secondary research and the IOM interview, we synthesized what we learned into a clear problem statement:

Communities in displacement settings are being asked to participate in carbon markets without the information, tools, or power to verify they are being treated fairly. Even when projects succeed technically, lack of transparency leaves communities unable to confirm fair benefit-sharing. And when projects fail due to developer integrity issues—as happened in IOM's pilot, communities who did everything right receive nothing and do not even understand why.

This became the foundation for everything that followed.

Phase 3: Ideate - Generating Solutions

With a clear problem in hand, we moved to brainstorming solutions. We generated dozens of ideas:

- A mobile app showing real-time carbon prices
- A blockchain-based transparent ledger
- A community-owned carbon cooperative
- A radio-only education campaign

- An independent auditor paid by communities
- A transparent intermediary that buys and sells credits
- A transparency layer that works with existing developers

We evaluated each idea against criteria: feasibility in low-connectivity contexts, alignment with IOM's model, trust-building potential, scalability, and cost.

Two ideas scored highest:

1. Education - communities need to understand carbon markets before they can hold anyone accountable
2. Transparency infrastructure - public price boards, itemized receipts, and community oversight

The insight: These are not separate solutions. They are two halves of one system. Education without transparency is just awareness. Transparency without education is overwhelming data. Together, they create informed accountability.

This is how Carbon Clarity was born: as an integrated transparency and education layer that works within existing carbon frameworks.

Phase 4: Prototype - Designing the Tools

Once we had the concept, we moved to designing specific tools:

- Visual VSLA training modules
- Household guides with pictures
- Public price board templates
- Itemized receipt formats
- Simple visual contracts
- VSLA Carbon Liaison role descriptions
- Benefit Fund Committee governance structures
- Grievance mechanism flowcharts

These are low-fidelity prototypes, sketches and templates, not finished products. But they are concrete enough to show IOM what "transparency infrastructure" actually looks like in practice.

Phase 5: Test - Validating with Stakeholders

We tested our concept in three ways:

First, with judges in the all-teams presentation. Their feedback told us:

- Phased rollout is essential to avoid overwhelming communities (Kamleshan)
- We need deeper technical understanding of the carbon lifecycle (Mercy)
- We need concrete examples of grievance mechanisms (Clara Maria)

We incorporated all three pieces of feedback into this final report.

Second, with IOM Mozambique through the interview. Mercy's responses validated the problem, corrected our positioning, and confirmed partnership feasibility.

Third, through literature validation. Every design choice we made is backed by research on what works in low-literacy, displacement contexts: VSLAs as governance structures, visual tools for low-literacy users, multi-channel communication (VSLA + radio + guides), participatory oversight, and phased information rollout.

What This Methodology Gives Us

This process design thinking grounded in real research, gives us three things:

Confidence. We are not guessing. Every element of Carbon Clarity responds to documented problems and validated needs.

Credibility. We can show judges, IOM, and funders exactly where our design came from and why we made each choice.

Adaptability. Because we built the solution through listening, we can iterate based on field feedback without losing the core logic.

SECTION 5: SOLUTION DESIGN

Building Transparency Into the System, Not Around It

What Carbon Clarity Actually Is

Before we explain how Carbon Clarity works, we need to be clear about what it is—and what it is not.

Carbon Clarity is not a carbon project developer. We do not register projects with Gold Standard or Verra. We do not design monitoring systems or conduct third-party verification. We do not issue carbon credits or sell them to buyers. That is not our role, and trying to do those things would duplicate what IOM and their partners already have in place.

Carbon Clarity is not a replacement intermediary. In our initial design, we thought we would buy credits from communities and sell them to buyers, operating as a transparent alternative to exploitative middlemen. But our research with IOM showed us this was the wrong approach. Carbon pricing mechanisms are already defined by carbon standards. Creating a parallel buying-and-selling system would conflict with those frameworks and make us just another layer communities have to trust.

So what is Carbon Clarity?

Carbon Clarity is a transparency and education infrastructure that works inside the existing carbon system to ensure communities understand the process and can verify they are treated fairly at every stage.

Think of it this way: Gold Standard and Verra provide the rules for how carbon projects should work. Carbon developers provide the technical capacity to execute those rules—registration, monitoring, verification, credit sales. IOM provides safeguarding, community access, and humanitarian oversight.

Carbon Clarity provides the missing piece: the tools and structures that make the developer's promises transparent and enforceable from the community's perspective.

We are not adding a new layer on top of the system. We are making the existing system work the way it is supposed to—with communities informed, empowered, and able to hold actors accountable.

Two Components Working as One System

Carbon Clarity has two integrated components. They are not separate tools—they are designed to reinforce each other at every stage.

Component 1: The Carbon Clarity Hub (Education & Communication)

This is how we build carbon literacy in communities. It includes:

- VSLA training modules that explain carbon markets in simple, visual language
- Household guides families can keep and refer to at home
- Community radio programs that repeat key messages and answer common questions
- Public price boards posted in community spaces showing current pricing, costs, and margins

The Hub does not just educate once and walk away. It builds understanding over time, starting simple and adding complexity only as communities demonstrate readiness and request more detail.

Component 2: Transparency Tools (Verification & Accountability)

This is how we make the carbon process visible and checkable. It includes:

- Itemized receipts with every payment showing price per credit, costs, margins, and shares
- Simple visual contracts in local language that communities actually understand before signing
- VSLA Carbon Liaison system where elected representatives review price boards and bring member concerns forward
- Benefit Fund Committees that co-design benefit-sharing formulas and can renegotiate if needed
- Clear grievance mechanisms with documented response times and public outcomes

The transparency tools do not work if communities lack the literacy to use them. That is why education comes first. But education alone does not create accountability—you need enforceable structures. Together, these two components create informed accountability.

How We Fit Into the Carbon Project Structure

Here is how a typical cookstove carbon project works, and where Carbon Clarity fits:

GOLD STANDARD / VERRA FRAMEWORKS
(Rules and requirements)
↓
IOM MOZAMBIQUE

(Facilitator, safeguarding, community access)
↓
CARBON PROJECT DEVELOPER
(Technical: registration, MRV, credit sales)
↓
CARBON CLARITY ← WE FIT HERE
(Transparency & education layer)
↓
COMMUNITIES VIA VSLAs
(Informed participants who can verify fairness)

The developer does their job: They register the project, set up monitoring systems, collect data on stove usage, hire third-party verifiers, get credits issued by the registry, and sell those credits to buyers. None of that changes.

Carbon Clarity makes that job transparent: We ensure communities understand what is happening at each stage, receive accessible updates on progress, and can verify pricing and benefit-sharing when revenues flow. We provide the tools and training that turn communities from passive beneficiaries into informed participants.

IOM provides the accountability bridge: Because IOM has relationships with both the developer and the communities, they can enforce transparency requirements. If a developer refuses to publish prices or provide itemized receipts, IOM can say: "Then you do not meet our partnership standards." This gives our transparency infrastructure teeth.

The Three-Year Timeline: Why Phased Implementation Matters

One of the clearest pieces of feedback we received from judges was this: do not overwhelm communities with all the information at once. Kamleshan put it directly: *"When a community sees a lot of information at one time, it does add for transparency, but they may be overwhelmed or inundated."*

He is right. And here is why the carbon timeline actually helps us design a better phased approach.

Mercy explained that carbon cookstove projects take 2-3 years from the start of deployment to when credits are sold and revenues flow. During that time, four major stages happen:

1. Project registration (Months 0-12): Developer submits project to Gold Standard/Verra for approval

2. Stove deployment and monitoring (Months 6-24): Stoves distributed, field staff track usage
3. Verification and credit issuance (Months 24-30): Third-party verifier checks data, registry issues credits
4. Credit sales and payment (Months 30-36+): Developer sells credits to buyers, revenues flow to communities

Most carbon projects wait until Stage 4 to think about community transparency. By then, it is too late. Communities are signing contracts they do not understand in Year 1, and only learning what those contracts mean in Year 3 when money appears—or does not appear—on their phones.

Carbon Clarity engages from Day 1, but phases the complexity:

PHASE 1: FOUNDATION (Months 0-6)

What is happening in the carbon project:

- IOM is distributing 2,770 cookstoves across nine sites
- Developer is finalizing project registration documents
- Baseline data is being collected (how much firewood households currently use)

What Carbon Clarity does:

- Launch basic VSLA training (2-3 sessions per group) covering:
  - What are carbon credits? (Simple answer: "Your stove reduces smoke and saves trees. That has value in carbon markets.")
  - How long does the process take? (2-3 years)
  - When should you expect money? (Not soon—only after credits are issued and sold)
- Distribute simple 4-page household visual guides showing: stove → emissions saved → carbon value → future payment
- Install one basic price board in each site showing: "Credits not yet issued. Project in registration phase. Expected timeline: 2-3 years."

Why this phase matters:
Communities need to understand upfront that carbon finance is slow and complicated. If they expect money next month and it takes three years, trust collapses. Phase 1 sets realistic expectations and builds basic carbon literacy without overwhelming people with details they do not need yet.

PHASE 2: BUILDING UNDERSTANDING (Months 6-24)

What is happening in the carbon project:

- Stoves are being used daily
- Field staff visit to check usage rates and collect monitoring data
- Developer prepares for third-party verification
- What Carbon Clarity does:
- Launch community radio programs (15 minutes weekly) that:
    - Explain what field staff are doing and why (MRV = monitoring, reporting, verification)
    - Answer common questions ("Why does it take so long?" "How do they calculate credits?")
    - Share updates on progress ("We are now in the verification stage")
- Expand VSLA training to cover:
    - What are costs? (Transport, verification fees, registry costs)
    - What is a margin? (What the developer keeps to cover their work)
    - What should benefit-sharing look like? (International standards say communities should get majority)
- Establish VSLA Carbon Liaison system:
    - Each VSLA elects one person to be the carbon point person
    - This person attends Carbon Clarity update sessions every quarter
    - Brings back information and member questions
- Update price boards quarterly: "Verification complete. Credits expected in Month 28."

Why this phase matters:
This is when communities start asking harder questions. They see field staff visiting and want to know what that means. They hear rumors about money and want to know when it is coming. Phase 2 builds deeper literacy so that when credits are issued and sold, communities are ready to understand the pricing breakdown, not seeing it for the first time.

PHASE 3: FULL TRANSPARENCY (Months 24-36+)

What is happening in the carbon project:

- Credits have been issued by the registry
- Developer is negotiating with buyers or has already sold credits
- Revenue is flowing
- What Carbon Clarity does:
- Activate full transparency tools:
    - Price boards show real numbers: "Market price: $12/credit. Your household: 2 credits = $24 value. Costs: $7. Margin: $3. Your payment: $10. Community fund: $4."
    - Itemized receipts distributed with every payment (paper or SMS) showing the full breakdown

- ➢ Benefit Fund Committees meet to review total revenues, approve community fund projects, and verify developer is following the agreed formula

❖ Launch full grievance mechanism:
- ➢ VSLA Carbon Liaisons can raise payment disputes
- ➢ Hotline for households to call if receipt seems wrong
- ➢ 2-week response guarantee with public outcomes

❖ Expand radio programs to include:
- ➢ Real examples: "Here is how Amina's payment was calculated. Here is how to check yours."
- ➢ Testimonials: "I compared my receipt to the price board and the numbers matched."

Why this phase matters:
This is when transparency has to deliver. If communities cannot verify the numbers now, after three years of preparation, the whole system fails. But because we built literacy in Phases 1 and 2, communities are not starting from zero—they have been learning for years and are ready to use the tools.

The Logic of Phased Rollout:

Phase 1 is about expectation management: Prevent disappointment by explaining upfront that this is a long process.

Phase 2 is about building capacity: Deepen understanding so communities can ask smart questions and spot problems.

Phase 3 is about enforcing accountability: Communities now have the knowledge and tools to verify fairness and challenge anything that seems wrong.

Kamleshan asked us to think about how information is received. This phased approach is our answer: we give communities what they need when they need it, building complexity over time rather than dumping it all at once.

Understanding the Technical Carbon Lifecycle - Where We Add Value

Another key piece of feedback came from Mercy, who asked us to demonstrate deeper understanding of how carbon markets actually work from project development through credit sales.

Here is the full technical lifecycle, with Carbon Clarity's role at each stage:

STAGE 1: PROJECT DEVELOPMENT & REGISTRATION (Months 0-12)

Who does it: Carbon project developer

What happens:

- Developer designs the project (how many stoves, where, what methodology)
- Submits Project Design Document (PDD) to Gold Standard or Verra
- Standard reviews and approves (or requests revisions)
- Project gets officially registered

Carbon Clarity's role:

- Communities do not need to understand all the technical details here, but they do need to know: "A project is being registered. This is the first step. It takes about a year."
- We provide simple updates via VSLA: "Registration submitted. Waiting for approval. No credits exist yet."
- Public price boards show: "Stage 1: Registration. Expected completion: Month 12."

Why this matters: Communities need to know the project is real and moving forward, even though no money flows yet. Without this, rumors start ("The project failed") or expectations spike ("Why haven't we been paid yet?").

STAGE 2: BASELINE & DEPLOYMENT (Months 0-18)

Who does it: IOM + partners + developer

What happens:

- ❖ Before stoves are distributed, developer collects baseline data: How much firewood do households currently use? How much do they spend on fuel?
- ❖ Stoves are distributed to 2,770 households
- ❖ Training on stove use and maintenance

Carbon Clarity's role:

- ❖ Phase 1 education happens here: What are carbon credits? Why does your stove create carbon value?
- ❖ Visual guides distributed with stoves so families have reference material at home
- ❖ VSLA training launches: "Your stove reduces emissions. That will eventually be worth money. Here is how the process works."

Why this matters: If we wait until credits are sold to educate communities, they have already been using stoves for 3 years without understanding why monitoring matters. Early education makes them better project participants.

STAGE 3: MONITORING (MRV Phase 1) (Months 12-24)

Who does it: Carbon project developer + field staff

What happens:

- Field staff visit households to check: Is the stove still being used? How often? Any problems?
- Usage data is logged (surveys, sensors, or "stove use monitors" - small devices that track temperature)
- Data shows: X households using stoves Y times per week = Z tonnes of wood saved

Carbon Clarity's role:

- Phase 2 education deepens here: "Field staff visits are part of MRV (monitoring, reporting, verification). This data proves your emissions reductions are real."
- Radio programs answer: "Why do they need to visit? What are they checking? Is my data private?"
- Price boards update: "Stage 3: Monitoring underway. Data collected from 2,500 households. Still no credits issued."

Why this matters: Monitoring can feel invasive if communities do not understand its purpose. Transparent communication makes it feel like partnership, not surveillance.

STAGE 4: REPORTING & VERIFICATION (Months 24-30)

Who does it: Developer prepares report → Third-party verifier audits

What happens:

- Developer writes Monitoring Report showing: X tonnes of CO_2 reduced across Y households over Z period
- Independent verifier (hired by developer, approved by Gold Standard/Verra) checks the data: Did monitoring follow the methodology? Is the data credible?
- Verifier issues Verification Statement: "We confirm X credits should be issued."

Carbon Clarity's role:

- Continue Phase 2 education: "A third-party expert is checking the data now. This is to make sure the numbers are accurate. Expected completion: Month 28."

- Price boards update: "Stage 4: Verification in progress. Third-party auditor reviewing data."

Why this matters: This stage is invisible to communities in most projects. They do not know verification is happening or why. Transparent updates build trust that the process is rigorous and real.

STAGE 5: CREDIT ISSUANCE (Month 30)

Who does it: Carbon registry (Gold Standard or Verra)

What happens:

- Registry reviews verification report
- Issues credits into developer's account in the registry system
- Credits now exist and can be sold

Carbon Clarity's role:

- Phase 3 activation begins: "Credits issued! The registry has confirmed X credits from your community."
- Price boards update with real numbers: "Credits issued: 4,000 tCO_2e. Market reference price: $12/credit. Estimated community share (formula): $28,800."
- Radio announces: "Credits are real now. Next step: developer sells them to buyers. Payment expected within 6 months."

Why this matters: This is the moment communities have been waiting 2.5 years for. If we have done our job in Phases 1 and 2, they understand what this means and what to expect next.

STAGE 6: CREDIT SALES & PAYMENT (Months 30-36+)

Who does it: Carbon project developer sells to buyers → Revenue flows

What happens:

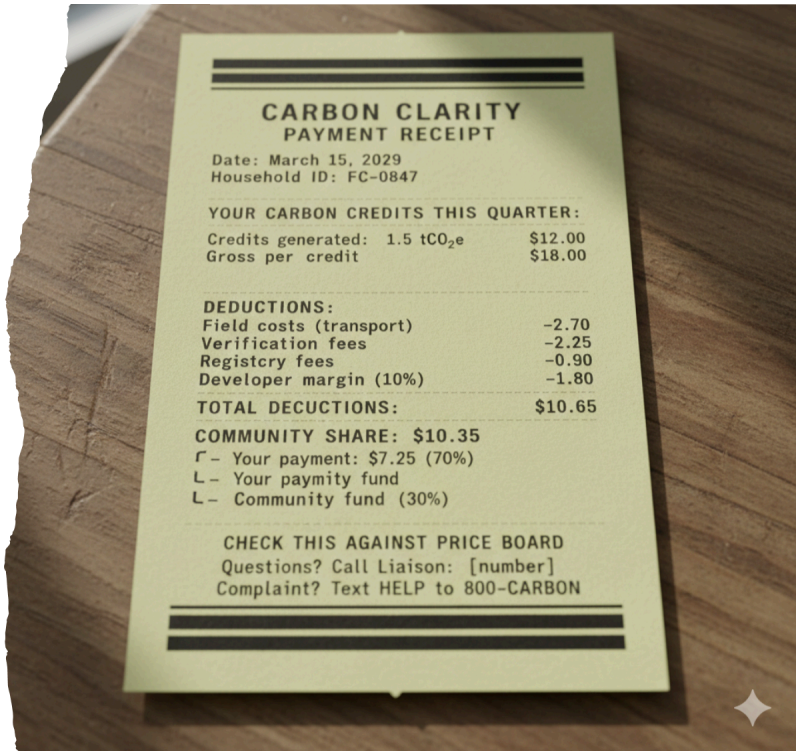
- Developer negotiates with corporate buyers, carbon funds, or brokers
- Credits are sold at agreed price (could be $8-15 per credit depending on quality, co-benefits, buyer demand)
- Developer receives payment from buyer
- Developer deducts their costs (transport, MRV, verification fees, registry costs, staff time, margin)
- Remaining revenue is distributed according to benefit-sharing agreement

Carbon Clarity's role - **THIS IS WHERE FULL TRANSPARENCY ACTIVATES:**

Price boards now show complete breakdown:



Itemized receipts distributed to each household:



VSLA Carbon Liaisons:

- Meet with Carbon Clarity team to review total revenues
- Compare receipts from 5-10 households against price board
- Raise any discrepancies immediately
- Report back to VSLA members: "Numbers checked, payments verified"

Benefit Fund Committee:

- Reviews total fund deposit: $8,280
- Meets with community to decide: Water project? School supplies? Health clinic support?
- Approves project, tracks spending, reports publicly

Grievance mechanism operational:

- If Amina's receipt shows $5 but she expected $7 based on price board, she tells her VSLA Carbon Liaison
- Liaison logs complaint, we investigate within 2 weeks
- Outcome posted publicly: "Error found and corrected" OR "Calculation explained: Amina's stove usage was lower this quarter (1.2 credits vs. expected 1.5)"

Why this stage matters:
This is the moment of truth. After three years of education and preparation, communities can now verify if promises were kept. The price board shows what was promised. The receipt shows what was delivered. If numbers do not match, there are clear channels to challenge. This is transparency enforced, not just promised.

Concrete Tools - What Communities Can Touch and Use

Let us move from process to artifacts. Here are the actual tools communities will have in their hands:

TOOL 1: VISUAL HOUSEHOLD GUIDE (4-8 pages)

Page 1: Cover with picture of woman cooking with improved stove, title in Emakhuwa: "Your Stove Creates Value - Here's How"

Page 2-3: Visual story with minimal text:

- Picture: Traditional 3-stone fire → lots of smoke, lots of wood
- Picture: Improved stove → less smoke, less wood
- Picture: Trees saved + cleaner air = carbon value
- Picture: Money from carbon buyers → you + community fund

Page 4-5: Timeline with pictures:

- Year 1: Stove distributed, project registered
- Year 2: Monitoring happens, data collected
- Year 3: Credits issued, payments start
- Simple message: "This takes time. Be patient."

Page 6-7: What to expect when payment comes:

- Picture of mobile phone with money + receipt
- Simple breakdown showing: Total → Costs → Margin → Your share → Fund share
- Picture of price board: "Check this to verify your payment"

Page 8: Who to contact:

- Picture of VSLA meeting: "Ask your Carbon Liaison"
- Picture of phone: "Call hotline if something is wrong"
- Picture of community board: "Check here every 3 months"

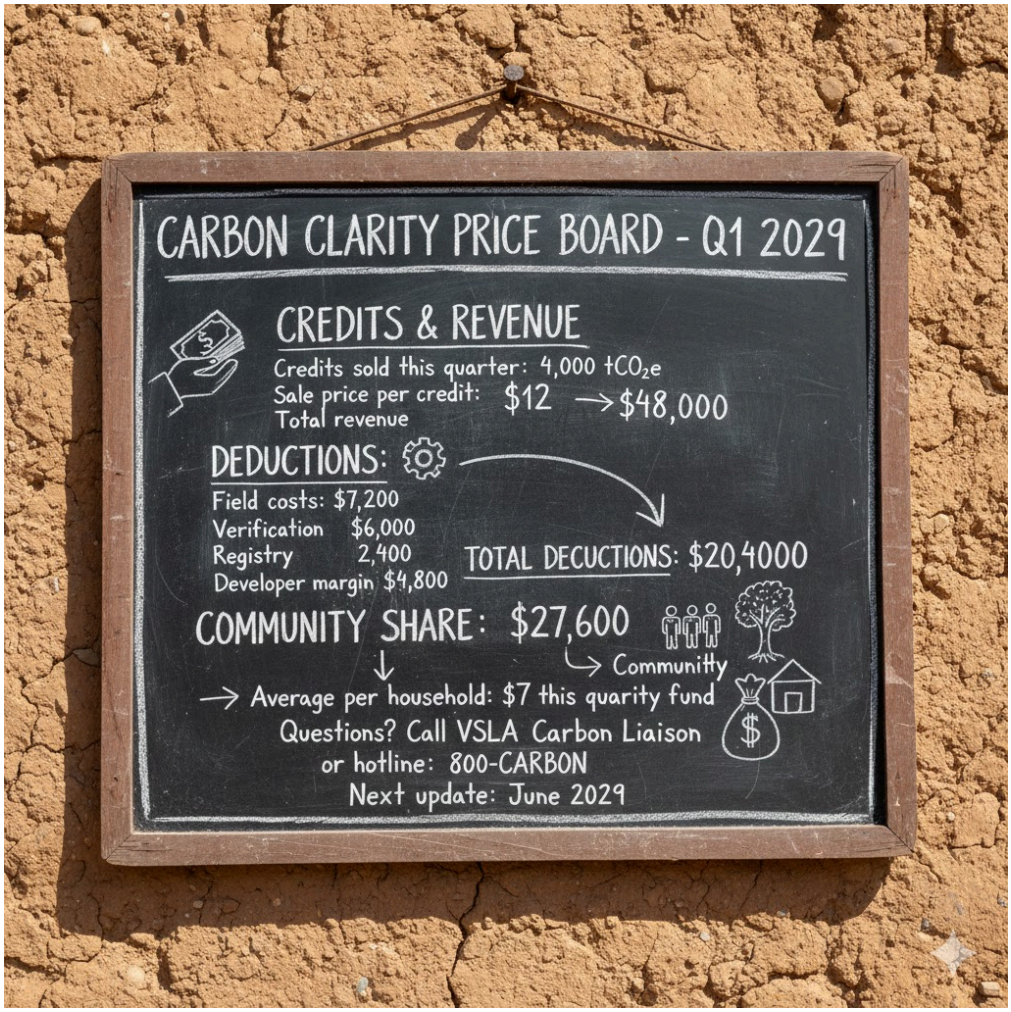
Design principles:

- ❖ More pictures than words
- ❖ Each sentence maximum 10 words
- ❖ Large text (24pt minimum)
- ❖ High contrast colors (works in bright sun)
- ❖ Laminated or durable paper (survives multiple users)

TOOL 2: PUBLIC PRICE BOARD (Durable, Weather-Resistant)

Mounted in: Central VSLA meeting space, visible to all, protected from rain

What it shows (updated quarterly):



Why this works: Anyone walking past can see the numbers. VSLA Carbon Liaisons reference it in meetings. Households can compare their receipt to board. Public visibility prevents hidden deals.

TOOL 3: VSLA TRAINING MODULE (3 sessions, 60 minutes each)

Session 1: Carbon Basics (Phase 1)

- What are carbon credits? (Simple definition with pictures)
- Why do cookstoves create carbon value? (Less wood = fewer trees cut = carbon stored)
- How long does it take? (2-3 years, here's why)
- Role-play: "Someone offers to buy your credits today for cash. Should you accept?" (No - credits don't exist yet, they're lying)

Session 2: Understanding Costs and Margins (Phase 2)

- What are legitimate costs? (Transport, verification, registry fees)
- What is a margin? (What the developer keeps for their work)
- What is fair? (International standards say communities should get 50%+)
- Activity: Practice reading a sample receipt, compare to price board

Session 3: Your Rights and How to Use Them (Phase 3)

- You have the right to see the numbers
- You have the right to ask questions
- You have the right to complain if something seems wrong
- Role-play: "Your payment seems low. What do you do?" (Tell Carbon Liaison, check receipt against board, file grievance if needed)

Format:

- Delivered by trained VSLA facilitators (not outsiders)
- Uses flipcharts with pictures
- Interactive, not lecture-style
- Conducted in Emakhuwa
- Repeated every 6 months for new members

TOOL 4: GRIEVANCE MECHANISM (Clear Process, Fast Response)

STEP 1: IDENTIFY THE PROBLEM

Amina receives payment but thinks it's wrong. She has three entry points:

- Tell her VSLA Carbon Liaison at weekly meeting
- Call/SMS hotline: 800-CARBON
- Visit Carbon Clarity office during open hours (Wednesdays 9am-12pm)

STEP 2: LOG THE COMPLAINT

Liaison or hotline staff fills simple form:

- Name and household ID
- What is the problem? (Payment too low / No payment received / Don't understand receipt / Other)
- What does she expect? (Amount or explanation)
- Date complaint received

Complaint gets unique number (#047) and is logged in public register.

STEP 3: INVESTIGATE (Within 7 days)

Carbon Clarity team:

- Checks registry: How many credits were attributed to her household?
- Checks payment records: What was sent to her phone?
- Checks price board: What should she have received based on formula?
- Identifies: Error on our side / Calculation correct but confusing / Third-party issue (developer/registry)

STEP 4: RESPOND (Within 14 days)

Three possible outcomes:

Outcome A - Error Found:
"We made a mistake. Your receipt showed 1.2 credits but registry shows 1.5. We owe you $3.60. Payment sent today plus $1 apology bonus. Sorry for the error."

Outcome B - Calculation Correct:
"We reviewed your case. Your stove usage was monitored at 70% this quarter (vs. 90% average), so you received 1.2 credits instead of 1.5. Here's the monitoring data. Your payment of $5.25 is correct based on the formula."

Outcome C - External Issue:
"The registry made an error in credit attribution. We are following up with them. We expect resolution within 30 days and will update you weekly."

STEP 5: PUBLIC OUTCOME (Within 21 days)

- Outcome posted on price board (complaint #047 resolved)
- Announced at next VSLA meeting
- Added to quarterly transparency report

Safeguards built in:

- No retaliation - Amina stays in program even if complaint is unfounded
- Public process - outcomes are visible to all
- Time-bound - 2-week maximum response
- Accessible - multiple entry points (VSLA, hotline, office)

Governance Structures - Who Decides and How

Transparency tools only work if communities have power to act on the information. Here's how we distribute that power:

TIER 1: HOUSEHOLD LEVEL (Individual Agency)

Every household:

- Receives their own copy of visual guide
- Gets itemized receipt with every payment
- Can opt out of project anytime without penalty
- Can file grievance without going through gatekeepers

Power dynamic: Individual households are not powerless. They have information and channels.

TIER 2: VSLA CARBON LIAISON (Group Oversight)

Each VSLA (14 groups) elects one Carbon Liaison who:

- Attends quarterly Carbon Clarity update sessions
- Reviews price board on behalf of VSLA members
- Collects questions and complaints from members
- Reports back at VSLA meetings with summaries
- Can request additional documentation if numbers seem off

Selection: Elected by VSLA members, usually for 1-year term, can be re-elected

Training: Receives extra training on reading price boards, checking receipts, logging grievances

Power dynamic: VSLAs already have strong accountability culture (they track small loans carefully). Extending that to carbon oversight is natural.

TIER 3: BENEFIT FUND COMMITTEE (Community Decision-Making)

Composition:

- 5-7 elected members from across VSLAs
- Must include: minimum 2 youth (under 35), minimum 3 women, at least 1 person with disability (if feasible)
- Elected every 18 months

Responsibilities:

- Reviews total community fund deposits quarterly
- Proposes and approves community fund projects (water, education, health)
- Co-designs benefit-sharing formula (household % vs. fund %)
- Can question developer cost claims and request receipts
- Can propose changes to formula or margin cap
- Meets quarterly, minutes posted publicly

Decision-making:

- Proposals require majority vote

- Major changes (formula revision) require community-wide meeting with 60% approval
- Financial decisions audited annually by external accountant

Power dynamic: This committee holds real power. They control a collective fund, can renegotiate terms, and can demand accountability from developer and Carbon Clarity.

Concrete Examples - Accountability in Practice

Clara Maria asked for specific examples of how grievance and inclusion mechanisms work. Here they are:

EXAMPLE 1: AMINA'S GRIEVANCE (Already detailed above in Stage 6)

Showing the 5-step process from complaint to public resolution in 14-21 days.

Key takeaway: Process is accessible (multiple entry points), fast (2-week response), and public (outcome visible to all). No retaliation.

EXAMPLE 2: ENSURING YOUTH VOICE IN GOVERNANCE

Challenge: Youth briquette producers (ages 20-28) feel sidelined by older VSLA leaders who dominate Benefit Fund Committee.

Our inclusion design:

✅ Mandatory youth quota: Committee MUST have 2 members under 35. If youth candidates don't run, election is invalid and must be redone.

✅ Separate youth producer VSLAs: Youth making briquettes can form their own VSLA groups, elect their own Carbon Liaison, have dedicated voice.

✅ Youth-targeted communication:

- Radio programs air 6-8pm (when youth listen)
- Visual guides use young people in illustrations
- Training sessions happen on weekends (when youth are free)

✅ Alternative participation channels:

- Youth can SMS grievances (don't have to speak in front of elders)
- Youth committee members get mentorship from experienced VSLA leaders

✅ Monitoring for exclusion:

- We track grievances by age/gender
- If zero youth complaints in 6 months, that's a red flag (means they're not comfortable using system)
- We do targeted outreach: "Are you really satisfied, or do you feel you can't speak up?"

Result: Youth have guaranteed representation AND multiple ways to participate even if intimidated by traditional structures.

EXAMPLE 3: SUPPORTING A WOMAN WITH DISABILITY

Challenge: Fatima (32, deaf) cannot participate in oral VSLA training sessions or hear radio programs.

Adaptive design:

✅ Visual-first materials: All guides work without sound. Pictures tell the story.

✅ Written backup: VSLA facilitator writes key points on paper for Fatima to read during sessions.

✅ Trusted intermediary allowed: Fatima can bring her daughter (who knows sign language) to meetings to interpret.

✅ SMS-based grievance: Fatima can text complaints with photos of her receipt if needed.

✅ Direct check-ins: Facilitator asks Fatima directly (via writing): "Do you understand today's session? Yes/No/Need clarification?" Doesn't assume silence means understanding.

✅ Committee consideration: If Fatima wants to serve on Benefit Fund Committee, we provide written agendas in advance so she can prepare, and allow her daughter to interpret during meetings.

Result: Disability doesn't block participation—design adapts. Fatima has equal access to information and equal power to challenge if needed.

SECTION 6: BUSINESS MODEL

Making Transparency Financially Viable

Why the Business Model Matters

A solution that depends on endless charity is not a solution—it is a temporary program waiting to collapse when the grant money runs out. Carbon Clarity must be financially sustainable. But here is the tension: we exist to protect communities from exploitation, which means we cannot become another extractive layer that captures value meant for households.

So our business model has to do two things at once:

- Generate enough revenue to cover operating costs and sustain long-term operations
- Ensure the vast majority of carbon value still reaches communities

This is not easy. But it is possible if we design carefully.

Understanding the Partnership Structure First

Before we talk about money, we need to be clear about who does what, because that determines who pays for what.

Here is the partnership model for IOM Mozambique's cookstove carbon project:

IOM MOZAMBIQUE:

- Distributes 2,770 cookstoves across nine sites
- Provides community access and safeguarding
- Ensures activities meet humanitarian standards
- Connects project to government and donors
- Does NOT handle carbon credits or registration

CARBON PROJECT DEVELOPER:

- Registers project with Gold Standard or Verra
- Designs and implements MRV (monitoring, reporting, verification) systems
- Hires third-party verifiers
- Gets credits issued by registry
- Sells credits to buyers
- Receives revenue from credit sales
- Distributes benefit-sharing payments to communities

CARBON CLARITY (US):

- Provides transparency and education infrastructure
- Trains VSLAs on carbon literacy
- Produces visual guides, price boards, receipts
- Establishes VSLA Carbon Liaison system
- Operates Benefit Fund Committee governance
- Runs grievance mechanisms
- Ensures developer's promises are transparent and enforceable

Key insight: We are not in the carbon business. We are in the transparency business. The developer sells credits. We make sure communities understand and can verify what happens with the revenue.

This partnership structure determines our revenue model.

Three Revenue Streams

Carbon Clarity operates on three revenue streams, designed to balance sustainability with fairness:

REVENUE STREAM 1: SERVICE FEE FROM PROJECT DEVELOPER (40-50% of operating costs)

What it is:
The carbon project developer pays Carbon Clarity a service fee for providing community education, transparency infrastructure, and grievance mechanisms.

Why developers pay this:
Three reasons:

First, it reduces their risk. Mercy told us that "low trust or poor communication" is one of the main reasons carbon projects fail. When communities do not understand what is happening, they disengage, rumors spread, and projects collapse. By paying Carbon Clarity to handle education and transparency, developers reduce the risk of project failure.

Second, it increases credit value. Carbon buyers increasingly demand proof of community benefit and consent. Credits from projects with transparent benefit-sharing and documented community oversight can command premium prices—sometimes $2-5 more per credit than opaque alternatives. Developers can justify paying for transparency because it increases their revenue.

Third, IOM can require it. IOM has told us they want transparency infrastructure in their scale-up phase. They can make Carbon Clarity's services a condition of partnership: "If you want to work with IOM's communities, you must provide transparent pricing and education. You can hire Carbon Clarity to do that, or do it yourself to the same standard." Most developers will choose to pay the fee rather than build the capacity in-house.

How much:
We estimate $15,000-20,000 per year for a project covering 2,770 households. That is roughly $5-7 per household—a small fraction of the $50-200 in carbon value each household generates over the project lifetime.

What it covers:

- VSLA facilitator training and ongoing support
- Production and distribution of visual guides
- Price board design, installation, and quarterly updates
- VSLA Carbon Liaison training and meetings
- Grievance mechanism operations

REVENUE STREAM 2: GRANT FUNDING FOR EDUCATION COMPONENT (30-40% of operating costs)

What it is:
Philanthropic foundations, climate funds, or impact investors provide grant funding specifically for the Carbon Clarity Hub education component.

Why funders support this:
Carbon literacy in vulnerable communities is a public good. It benefits not just one project, but builds capacity that communities can use to evaluate any future carbon opportunity. Funders increasingly want to support infrastructure that prevents exploitation and builds community agency—exactly what Carbon Clarity provides.

Examples of potential funders:

- Climate finance capacity-building funds (GCF Readiness, CDKN)
- Humanitarian innovation funds (IKEA Foundation, UNHCR Innovation Fund)
- Transparency and governance initiatives (Open Society, Hewlett Foundation)
- Corporate social responsibility programs from carbon buyers

How much:
We estimate $10,000-15,000 per year for the education component across nine sites.

What it covers:

- Radio program production and airtime
- Advanced training for Benefit Fund Committees
- Monitoring and evaluation of community understanding
- Continuous improvement of education materials

Sustainability pathway:
Grant funding is needed most in the first 3-5 years while the model is being proven. Once Carbon Clarity demonstrates impact—communities are informed, transparency works, projects succeed—the service fee model can cover more of the education costs because developers see the value clearly.

REVENUE STREAM 3: RESULTS-BASED FINANCE (10-20% of operating costs, grows over time)

What it is:
Impact investors or outcome funders pay Carbon Clarity based on verified results: number of households demonstrating carbon literacy, grievances resolved fairly, communities reporting trust in the system.

Why this matters:
This ties our revenue directly to impact. If communities do not understand carbon markets or

cannot verify fair treatment, we do not get paid. This aligns our incentives with community outcomes, not just activity counts.

Example structure:

- $2 per household that passes a basic carbon literacy assessment (can explain what carbon credits are, where to check prices, how to file grievance)
- $5 per grievance resolved within 2-week timeframe with documented community satisfaction
- $10 per household reporting "high trust" in carbon transparency mechanisms (via annual survey)

How much:
For 2,770 households, if 70% pass literacy assessment over first 2 years, that is ~$3,900. Add grievance resolution and trust metrics, total potential: $8,000-12,000 over 3 years.

What it covers:

- Quality assurance and continuous improvement
- Independent verification of impact
- Documentation of lessons learned for replication

Total Operating Budget - Three-Year Pilot

Here is what it costs to run Carbon Clarity for a 2,770-household project over three years:

YEAR 1 (Phase 1: Foundation)

- VSLA facilitator training: $8,000
- Visual guide production: $5,000
- Price board design & installation: $4,000
- Staff (1 coordinator, 2 field officers): $18,000
- Transport & logistics: $6,000
- Office & admin: $4,000
  Year 1 Total: $45,000

YEAR 2 (Phase 2: Building Literacy)

- Radio program production & airtime: $8,000
- VSLA Carbon Liaison training: $4,000
- Benefit Fund Committee setup: $3,000
- Staff: $20,000
- Transport & logistics: $7,000

- Office & admin: $4,000
  Year 2 Total: $46,000

YEAR 3 (Phase 3: Full Transparency)

- Grievance mechanism operations: $6,000
- Receipt production & distribution: $4,000
- Quarterly price board updates: $2,000
- Benefit Fund Committee support: $3,000
- Staff: $22,000
- Transport & logistics: $7,000
- Office, admin & evaluation: $6,000
  Year 3 Total: $50,000

TOTAL 3-YEAR PILOT: $141,000

Per household over 3 years: ~$51

Per household per year: ~$17

Revenue Coverage

Year 1:

- Developer service fee: $18,000 (40%)
- Grant funding: $20,000 (44%)
- Results-based (partial): $2,000 (4%)
- Gap requiring additional grant: $5,000 (11%)

Year 2:

- Developer service fee: $20,000 (43%)
- Grant funding: $18,000 (39%)
- Results-based: $4,000 (9%)
- Gap: $4,000 (9%)

Year 3:

- Developer service fee: $25,000 (50%)
- Grant funding: $15,000 (30%)
- Results-based: $6,000 (12%)
- Gap: $4,000 (8%)

Key trend: Service fee grows as developer sees value. Grant dependency decreases. Results-based revenue grows as impact is proven.

By Year 4-5: Model should be 60-70% covered by service fees, 20-30% by grants, 10% by results-based finance.

Why This Is Fair and Sustainable

Fair because:

1. Communities pay nothing. Carbon Clarity's costs come from developers and funders, never from household carbon payments. The $17/household/year cost is covered entirely by external sources.
2. Developer fee is reasonable. $20,000/year for 2,770 households is 0.3-0.5% of total expected carbon revenue over project lifetime ($4-6 million at $10-15/credit). Transparency infrastructure that costs 0.5% of revenue and prevents project failure is a bargain.
3. We don't touch carbon payments. Unlike traditional intermediaries who take a cut of every credit sold, Carbon Clarity never handles carbon money. Developer sells credits, deducts costs and margin, pays communities directly. We get paid separately for our transparency service—not as a percentage of household payments.

Sustainable because:

1. Multiple revenue streams. We are not dependent on one funder or one partner. If grants decrease, service fees can increase. If one developer exits, others see the proven model and want to replicate.
2. Value is demonstrable. After 3 years, we can show: X households educated, Y grievances resolved, Z% community trust, zero project failures due to opacity. That evidence attracts more funders and justifies higher service fees.
3. Designed for replication. Once the model works in Mozambique, it can be replicated in other countries, other carbon activities (agroforestry, REDD+, renewable energy). Each replication increases revenue while per-unit costs decrease.

What We Do NOT Do (And Why That Matters for the Business Model)

It is important to be clear about what Carbon Clarity does NOT charge for or take money from:

We do NOT:

- Take a percentage of household carbon payments
- Handle carbon credit registration or sales
- Charge communities for training, guides, or grievance services

- Require communities to use our services (it is provided automatically as part of project)
- Take a margin on community fund deposits

Why this matters:
Our business model is designed so we never have an incentive to reduce what communities receive. A traditional intermediary makes more money when they pay communities less. We make the same service fee regardless of what communities are paid, so our only incentive is to ensure communities get maximum fair value, because that builds trust and makes the project (and our model) successful.

The Value Proposition for Each Actor

For communities:

- Free education and transparency infrastructure
- No reduction in their carbon payments to cover our costs
- Clear verification that they are receiving fair share

For developers:

- Reduced project risk (better community engagement = fewer failures)
- Higher credit value (transparent projects command premium prices)
- Outsourced community education and safeguarding (saves them building this capacity)

For IOM:

- Can engage in carbon finance without compromising humanitarian principles
- Documented community benefit and protection
- Replicable model for other displacement contexts

For funders:

- Investment in transparency infrastructure that prevents exploitation
- Measurable impact (literacy rates, grievance resolution, trust metrics)
- Contribution to broader carbon market integrity in Africa

For buyers:

- Documented community consent and benefit
- Lower reputational risk
- Alignment with high-integrity standards (ICVCM, ACMI)

Everyone benefits—except those who profit from opacity.

Financial Sustainability Beyond the Pilot

After the 3-year pilot in Mozambique, what happens?

Pathway 1: Scale within Mozambique

- IOM expands to more sites (additional 5,000-10,000 households)
- Carbon Clarity replicates model with same partnership structure
- Per-household costs decrease due to economies of scale (training materials already developed, systems in place)
- Service fee model becomes primary revenue source

Pathway 2: Replicate in other countries

- UNHCR, IOM offices in Kenya, Uganda, DRC, Malawi want similar model
- Carbon Clarity provides franchise/partnership: we train local NGOs to deliver, they pay licensing fee
- Reduces our direct operational costs, increases impact

Pathway 3: Expand to other carbon activities

- Model works for any community carbon activity: agroforestry, REDD+, renewable energy, sustainable agriculture
- Same transparency tools, adapted to different activities
- Developers in those sectors see proven model, want to replicate

Pathway 4: Become industry standard

- Gold Standard or Verra incorporates our transparency requirements into certification
- All projects must provide itemized receipts, public price boards, VSLA-linked oversight
- Carbon Clarity becomes training/certification provider for projects seeking high-integrity label

Long-term vision: Carbon Clarity becomes self-sustaining social enterprise that scales across Africa, making transparency the default rather than the exception in community carbon markets.

SECTION 7: GOVERNANCE & SAFEGUARDS

Power Cannot Be Promised - It Must Be Built Into the System

Why Governance Is Not a Side Feature

Here is a truth about transparency that many people miss: information alone does not create accountability. You can post all the numbers you want on a public board, but if communities have no structured way to question those numbers, no power to demand corrections, and no protection from retaliation when they raise concerns, then transparency is just theater.

This is why governance is not something we add on after designing the transparency tools. Governance is the foundation that makes transparency meaningful. Without it, price boards are just pretty posters. With it, they become evidence communities can use to hold powerful actors accountable.

Let us be direct about what we mean by governance: the formal structures and processes that distribute power, enable participation, ensure accountability, and protect the most vulnerable from being sidelined or harmed.

In Carbon Clarity, governance shows up in three places: who makes decisions, how those decisions get made, and what happens when someone with less power challenges someone with more power. If we get governance wrong, everything else fails.

The Problem Governance Must Solve

Before we describe our governance structures, we need to name the specific problems they are designed to prevent. Because governance is not abstract—it exists to stop real harms that happen in real carbon projects.

Problem 1: Information exists but communities cannot act on it.

Imagine price boards show that the developer is taking a 25% margin when the agreed cap was 15%. Communities see the discrepancy. But there is no clear process to challenge it, no designated person responsible for raising the issue, and no timeline for when the developer must respond. So the information sits there, visible but unenforceable. Communities feel frustrated, not empowered.

Our governance response: VSLA Carbon Liaisons are specifically responsible for reviewing price boards and raising discrepancies. Benefit Fund Committees have the formal authority to demand explanations and, if unsatisfied, escalate to IOM. This is not "anyone can complain"—it is "these specific people have the responsibility and power to challenge on behalf of the group."

Problem 2: Elite capture—powerful individuals control decisions meant for the whole community.

In many community programs, a small group of well-connected individuals (usually older men, local elites, or those with education) dominate committees and decision-making. They make choices that benefit themselves or their networks, not the broader community. Women, youth, people with disabilities, and marginalized groups get excluded—not explicitly, but through meeting times they cannot attend, language they do not speak, or social dynamics that silence them.

Our governance response: Mandatory representation quotas (minimum youth, minimum women, minimum from different VSLAs), staggered election timelines so no one group controls everything, public financial reporting so capture is visible, and alternative participation channels (SMS grievances, written feedback) so people who feel uncomfortable speaking in front of elites have other ways to raise concerns.

Problem 3: Retaliation—people who question or complain get punished or excluded.

If raising a grievance means you might get excluded from future payments, or face social backlash from project staff or community leaders, then grievance mechanisms are useless. People will stay silent even when they are being treated unfairly, because speaking up feels more dangerous than accepting the injustice.

Our governance response: Explicit no-retaliation policy written into contracts. Households who file grievances cannot be removed from the project—period. Grievance outcomes are public, which makes it visible if someone disappears after complaining. And VSLA Carbon Liaisons can raise issues on behalf of members anonymously, so individuals do not have to expose themselves.

Problem 4: Decisions happen too slowly or not at all.

Communities raise concerns and wait months for responses. Or committees meet once a year and cannot address urgent issues. Slow governance kills trust just as effectively as no governance.

Our governance response: Mandatory timelines (2-week grievance response, quarterly committee meetings, annual formula reviews), public tracking of open issues (posted on price boards), and escalation pathways if timelines are not met (from Carbon Liaison → Benefit Fund Committee → IOM oversight).

These four problems, powerless information, elite capture, retaliation, and slow response—are what our governance structures exist to prevent. Keep them in mind as we describe how governance actually works.

Three Tiers of Power Distribution

Carbon Clarity distributes decision-making power across three levels. Not because three is a magic number, but because communities need multiple entry points—individual agency, collective voice through trusted groups (VSLAs), and formal oversight structures (committees). Here is how they work:

TIER 1: HOUSEHOLD LEVEL - INDIVIDUAL AGENCY

What this tier controls:

Every individual household has the power to:

- Decide whether to participate in the carbon project at all
- Opt out at any time without penalty or explanation
- Receive their own copy of contracts, guides, and receipts
- File a grievance directly (not just through a representative)
- Keep or withdraw from VSLAs and still remain in the carbon project

Why this matters:

Individual agency is the baseline. If everything else fails—if VSLA leaders are corrupt, if committees are captured, if grievance mechanisms are broken—an individual household still has the nuclear option: walk away. That exit power, even if never used, creates pressure on other tiers to be responsive. Because if too many households exit, the project collapses.

How we protect it:

- Contracts explicitly state: "You can leave this project anytime by telling your VSLA Carbon Liaison or calling the hotline. No questions asked. If you have already generated credits that have not been paid, you will still receive payment for those."
- Opt-out process is simple: one phone call or one conversation. No paperwork, no waiting, no penalties.
- We track opt-out rates. If many households from one VSLA are leaving, that is a signal something is wrong in that group's governance—we investigate.

Concrete example:

Fatima feels her VSLA Carbon Liaison is not representing her concerns accurately. She does not trust him to raise her grievance about a low payment. Under Tier 1, Fatima does not have to go through him, she can call the hotline directly, file her grievance, and get a response within 2 weeks. Her individual agency is protected even when group structures are not working for her.

TIER 2: VSLA OVERSIGHT - COLLECTIVE VOICE THROUGH TRUSTED STRUCTURES

What this tier controls:

Each VSLA group (14 total in IOM's sites) elects one Carbon Liaison who:

- Represents the group at quarterly Carbon Clarity update sessions
- Reviews the public price board every month
- Collects questions and concerns from VSLA members
- Raises issues to Carbon Clarity, IOM, or the developer
- Reports back to VSLA meetings with updates and explanations
- Can request additional documentation (receipts, cost breakdowns, developer contracts)

Why this matters:

Individual households have power, but they may not have confidence or capacity to challenge a developer alone. VSLAs provide collective power. When a VSLA Carbon Liaison says "five of our members received lower payments this quarter and we want to understand why," that carries more weight than five individuals calling separately. And because VSLAs already exist and are trusted (members choose to save money together—they trust each other with cash), building carbon oversight into VSLAs leverages existing social capital rather than creating something from scratch that takes years to build trust.

How we protect it:

- Elections happen annually. If a Carbon Liaison is not serving members well, they can be replaced.
- Role is clearly defined: this is not "represent the developer to the community," it is "represent the community to the developer." Loyalty is to VSLA members.
- Training provided: Carbon Liaisons receive extra training on reading price boards, interpreting receipts, logging grievances. We do not assume they already have this skill—we build it.
- Compensation: Carbon Liaisons receive a small stipend ($5-10/quarter) from the community fund for their time, so this is not unpaid labor.

Concrete example:

Three households in one VSLA receive payments that seem inconsistent—one got $7, one got $5, one got $9, and they cannot figure out why. Instead of each filing individual grievances, they bring it to their Carbon Liaison at the weekly VSLA meeting. The Liaison reviews all three receipts, checks the price board, and realizes the difference is due to varying stove usage rates that quarter. She explains this at the next meeting, showing the monitoring data. Issue resolved without formal grievance, because the Liaison could access information and explain it in terms members understand. Trust maintained.

TIER 3: BENEFIT FUND COMMITTEE - FORMAL COMMUNITY GOVERNANCE

What this tier controls:

The Benefit Fund Committee (5-7 elected members) holds the most formal power:

Financial authority:

- Controls community benefit fund (20-30% of total community share)
- Approves or rejects proposed fund projects (water, education, health)
- Reviews fund spending quarterly with public reports

Oversight authority:

- Reviews total carbon revenues and cost deductions quarterly
- Can question developer cost claims and request supporting receipts
- Can propose changes to benefit-sharing formula (% to households vs. fund)
- Can propose changes to developer margin cap

Accountability authority:

- Receives and reviews grievance logs (how many complaints, response times, outcomes)
- Can escalate unresolved issues to IOM oversight
- Can recommend excluding a developer from future projects if integrity violations are found

Why this matters:

This tier holds the power that makes transparency enforceable, not just performative. If the committee reviews quarterly reports and sees costs are higher than justified, they can demand receipts. If receipts show the developer charged $10,000 for transport but comparable projects cost $4,000, they can challenge it and, if necessary, reduce the margin or renegotiate. This is not symbolic oversight—it is real financial and decision-making power.

How we protect it:

Inclusive composition (addressing Clara Maria's concern about vulnerable groups):

The committee MUST include:

- Minimum 3 women (out of 5-7 members) = 50%+ women's voices
- Minimum 2 youth under 35 = youth perspective required
- Representatives from at least 3 different VSLAs = no one VSLA dominates
- At least 1 person with disability (where feasible) = disability inclusion
- Mix of IDP and host community members = both groups represented

If an election produces a committee that does not meet these criteria, the election is invalid and must be redone. This is not optional.

Decision-making rules:

- Proposals require simple majority (3 out of 5 or 4 out of 7)
- Major changes (formula revision, margin cap reduction) require community-wide vote with 60% approval after public consultation
- Financial approvals over $1,000 require two committee members' signatures plus public notice 1 week before decision

Transparency and accountability:

- Committee meets quarterly (minimum)
- Minutes are written (even if brief) and posted publicly
- Financial reports show: fund balance, deposits this quarter, approvals made, current projects, spending to date
- Annual external audit by independent accountant (paid from fund admin budget)

Term limits:

- Members elected for 18-month terms
- Can be re-elected once (max 3 years continuous service)
- Then must step down to create space for others

Concrete example - Committee challenges developer costs:

The Benefit Fund Committee reviews Q3 financial report. Developer reports $8,000 in "field costs" for the quarter. Committee member (a youth VSLA leader) asks: "What does field costs include? That seems high." Carbon Clarity provides breakdown: $5,000 transport, $2,000 staff per diems, $1,000 materials. Committee compares to previous quarters and sees transport jumped from $3,000 to $5,000. They ask developer for fuel receipts and travel logs.

Developer provides documentation showing they visited 12 additional remote sites this quarter for expanded monitoring (hence higher transport). Committee accepts explanation but requests: "Please itemize transport separately from now on so we can track it." Developer agrees. In next quarter's report, transport is its own line item.

Result: Committee exercised oversight, developer was held accountable, transparency improved. Community trust maintained because people saw their representatives asking tough questions and getting answers.

Safeguards - Making Rights Actionable, Not Just Written

Governance structures create power. Safeguards ensure that power is used to protect people, especially the most vulnerable. Here are the specific safeguards built into Carbon Clarity:

SAFEGUARD 1: FREE, PRIOR, AND INFORMED CONSENT (FPIC)

What this means in practice:

FREE: Communities are not coerced or pressured into participating. They have real choice, including the choice to say no or to leave later.

PRIOR: Information and explanation come before decisions, not after. Contracts are explained in VSLA meetings before anyone is asked to sign.

INFORMED: Information is accessible—local language (Emakhuwa), visual formats, oral explanations, repeated multiple times. "Informed" does not mean "we handed them a 20-page document"—it means "they understand enough to make a real choice."

How we enforce it:

- No mass sign-ups. Contracts are explained group by group, with time for questions.
- Visual one-page summary accompanies every contract.
- Households can take contracts home, discuss with family, and return later to sign—no pressure to decide on the spot.
- Consent is ongoing: major changes to formula or terms require new rounds of explanation and re-consent.

SAFEGUARD 2: ACCESSIBLE INFORMATION FOR LOW-LITERACY CONTEXTS

The challenge:

Female literacy in Cabo Delgado is 30-35%. Most people speak Emakhuwa, not Portuguese. Standard carbon project documents are long, technical, and in English or Portuguese. This is not informed consent—it is signing blind.

Our response:

- All community-facing materials in Emakhuwa first, Portuguese second
- Visual guides with pictures that tell the story without requiring reading
- Oral explanations at every VSLA meeting (not just once)
- Radio programs in local language
- Price boards use large text, simple numbers, and icons
- Receipts use both words and simple graphics (pie chart showing your share vs. costs vs. margin)

Concrete example:

Amina received secondary education and can read Portuguese. Her neighbor Maria never attended school and cannot read any language. Both receive the same visual guide. Amina reads the text. Maria looks at the pictures and asks Amina to explain. At VSLA meeting, facilitator shows the same pictures on a flipchart and explains orally. Maria hears it again on the radio. By the time payments come, Maria may not be able to read the receipt, but she knows to check the big number at the bottom and compare it to what the price board showed. If they don't match, she

knows to tell the VSLA Carbon Liaison. She is informed enough to protect herself, even without literacy.

SAFEGUARD 3: GRIEVANCE WITHOUT RETALIATION

The fear:

"If I complain, they will cut me out of future payments. Better to stay quiet."

Our response:

Explicit no-retaliation clause in contracts:
"If you file a complaint or question a payment, you cannot be removed from this project or have future payments reduced. This is your right."

Public grievance log:
All grievances are logged with unique numbers (#001, #002, etc.) and posted publicly with status: "Under review," "Resolved," "Escalated." If someone files complaint #045 and then disappears from the project, that is visible to everyone—it triggers investigation.

Anonymous option through VSLA Liaison:
If Fatima is scared to put her name on a complaint, she can tell her Carbon Liaison: "Please raise this concern on behalf of a member who prefers to stay anonymous." Liaison logs it as "VSLA Group 7 - anonymous member query" and represents the issue without exposing Fatima.

Monitoring for retaliation:

We track: Did anyone leave the project within 3 months of filing a grievance? If yes, we contact them (via VSLA Liaison or directly) to ask why. If we find retaliation, the developer faces consequences—up to and including I

SAFEGUARD 3: GRIEVANCE WITHOUT RETALIATION (continued)

Monitoring for retaliation:

We track: Did anyone leave the project within 3 months of filing a grievance? If yes, we contact them (via VSLA Liaison or directly) to ask why. If we find retaliation, the developer faces consequences, up to and including IOM terminating the partnership and bringing in a different

developer. This is not a theoretical threat. IOM's humanitarian mandate means they cannot work with partners who punish communities for speaking up.

Concrete example:

Paulo filed a grievance in March claiming his payment was calculated incorrectly. In April, he receives a visit from a field officer who suggests "maybe it would be better if you left the project since you're not happy." Paulo feels pressured and tells his VSLA Carbon Liaison he is thinking of leaving.

The Liaison reports this to the Benefit Fund Committee. The committee raises it with Carbon Clarity and IOM. Within 48 hours, IOM's safeguarding team contacts the developer: "This appears to be retaliation. Explain what happened and ensure it stops immediately, or we terminate the partnership." Developer investigates, finds the field officer acted inappropriately, disciplines the officer, and sends a formal apology to Paulo. Paulo stays in the project. The incident is documented publicly.

Result: Communities see that retaliation is taken seriously and stopped. Fear decreases. More people feel safe raising concerns.

SAFEGUARD 4: INCLUSION OF WOMEN, YOUTH, AND PEOPLE WITH DISABILITIES

The challenge:

Clara Maria asked us directly: "How would you ensure that women, youth and the most vulnerable households can actually participate meaningfully in the community structures? How do you make sure they're not excluded?"

This is not a hypothetical concern. In many community programs, women attend meetings but do not speak. Youth are told "the elders will decide." People with disabilities cannot physically access meeting spaces or cannot participate because materials are not adapted.

Our multi-layered response:

Layer 1: Structural guarantees (quotas)

- Benefit Fund Committee: minimum 3 women, minimum 2 youth, at least 1 person with disability
- VSLA Carbon Liaisons: if 14 VSLAs elect 14 men, we require at least 5 to be women (we work with VSLAs to encourage female candidates)
- Meeting facilitation: we train female facilitators specifically to create space for women's voices

Why quotas matter: Without them, social dynamics default to existing power structures. Quotas are not perfect, but they force systems to create space for voices that would otherwise be drowned out.

Layer 2: Alternative participation channels

Not everyone is comfortable speaking in front of large groups or authority figures. So we create multiple ways to participate:

- SMS/text grievances: Youth and women can text concerns without speaking publicly
- Written feedback: Forms with checkboxes (☐ Payment too low ☐ Don't understand receipt ☐ Other) so people can communicate without full literacy
- Small group discussions: VSLA meetings (15-30 people) are less intimidating than community-wide meetings (200+ people)
- Anonymous submission: Drop-box for written concerns at VSLA meeting spaces

Layer 3: Adaptive design for disabilities

We provided concrete examples in Section 5, but let me add the reasoning:

People with disabilities are often the most excluded in community programs—not because of malice, but because programs are not designed with them in mind. We start with the assumption that if someone with a disability cannot participate, that is a design failure, not their problem. So we adapt:

- For deaf/hearing-impaired: Visual materials that work without sound, written summaries, allow sign language interpreters or family members to attend
- For blind/vision-impaired: Oral explanations at every session, allow trusted person to read receipts and price boards aloud, SMS receipts can be read by phone screen readers
- For mobility-impaired: Hold meetings in accessible locations, allow proxy attendance (send family member), provide home visits for receipt explanation if needed

Layer 4: Monitoring for exclusion

Here is the key insight: exclusion is often invisible because excluded people do not show up to complain. So we look for absence as a signal:

- If zero women file grievances in 6 months, we don't celebrate—we investigate: "Are women really satisfied, or do they not feel safe using the system?"
- If youth are 40% of the population but only 10% of the Benefit Fund Committee over multiple election cycles, we ask: "What barriers prevent youth from running or being elected?"
- If no one with disabilities ever serves on committees, we ask: "Are meetings accessible? Are we providing support?"

We track participation data by gender, age, disability status and look for patterns that signal exclusion. When we find them, we adapt.

Concrete example - Youth voice mechanism:

In one VSLA, youth feel older members dominate discussions and dismiss their concerns about briquette production not being included in carbon education materials. Youth are frustrated but don't speak up in meetings.

Carbon Clarity notices: youth attendance at VSLA meetings has dropped from 12 to 4 over three months. We do targeted outreach, facilitator meets separately with youth group and asks: "Why aren't you coming to meetings? What's not working?"

Youth explain the dynamic. We respond:

1. Create separate youth producer training sessions specifically on briquette carbon value
2. Invite youth to present at next VSLA meeting about briquette opportunities
3. Add one youth-only breakout discussion time to VSLA meetings (30 minutes where youth can talk separately, then report back)
4. Ensure one youth representative is elected to Benefit Fund Committee

Youth attendance increases. Youth concerns get heard. Older members learn from youth about briquette potential. Everyone benefits.

Result: Exclusion was spotted through attendance data, addressed through adaptive design, and resolved through creating space for youth voice.

SAFEGUARD 5: DO NO HARM AND CONFLICT SENSITIVITY

The challenge:

Mercy warned us explicitly: displacement contexts are fragile. Introducing new economic opportunities (carbon payments) or resources (community funds) can create tensions or worsen existing conflicts if not handled carefully.

Specific risks in northern Mozambique:

IDP vs. host community tensions: IDPs may be resented by host communities who feel they are competing for resources. If carbon payments only go to IDPs or are perceived as favoring IDPs, this can deepen tensions.

Gender dynamics: If men control carbon payments meant for women (who do the cooking), this can increase domestic tensions or reinforce inequalities.

Elite capture: If local elites (chiefs, politicians, wealthy families) control community funds, it can create resentment and undermine the project.

Ethnic or clan divisions: If benefit distribution is seen as favoring one ethnic group or clan over another, old conflicts can resurface.

Our conflict-sensitive design:

Principle 1: Inclusive targeting

Carbon payments do not favor IDP vs. host, anyone using a stove gets payment based on stove usage, not displacement status. Community fund projects must benefit both IDP and host areas (e.g., water point accessible to both).

Principle 2: Individual payments

Carbon payments go directly to the household registered as the stove user (usually the woman who cooks), via her personal mobile money account. Not through male household heads, not through community leaders. This reduces risk of money being diverted or controlled by others.

Principle 3: Transparent criteria

Any differences in payment (one household gets $7, another gets $5) must have clear, objective explanations visible to all (e.g., stove usage rates shown in monitoring data). This prevents rumors that differences are due to favoritism or discrimination.

Principle 4: Conflict monitoring

VSLA facilitators and Carbon Liaisons are trained to spot early warning signs:

- Are certain groups complaining about unfair treatment?
- Are people suddenly not attending meetings after being regular participants?
- Are there rumors spreading about the project causing division?

If tensions arise, we pause and address them before continuing. Better to slow down than to worsen conflict.

Principle 5: Local leadership involvement

We work with traditional leaders, religious leaders, and local government from the beginning, not as controllers of the project, but as informed stakeholders who can help mediate if tensions arise. Their buy-in and understanding prevents them from becoming spoilers.

Concrete example:

In one site, rumors start that IDP households are receiving higher carbon payments than host community households for the same stoves. Host community members feel this is unfair—"we welcomed them and now they get more money?"

VSLA Carbon Liaison brings the concern to Carbon Clarity and IOM. We investigate: Are payments actually different? We review data and find payments are identical—same stove type, same usage rate, same payment. The rumor was false.

But ignoring it would let tensions grow. So we:

1. Hold joint VSLA meeting with IDP and host members
2. Show the price board and sample receipts side-by-side
3. Explain: "Payments are based only on stove usage, nothing else. Here's the formula. Anyone can check."
4. Invite traditional leader to verify the data independently
5. Post clarification on price board: "All households with same stove type and usage receive same payment"

Rumor is dispelled. Tension defused. Trust restored through transparency and proactive communication.

Result: Conflict sensitivity means watching for tensions, addressing them quickly, and using transparency as a conflict prevention tool.

Why This Governance Model Is Different

Let us be honest: many carbon projects have "community committees" and "grievance mechanisms" on paper. They write it in project documents to satisfy donors and standards. But those structures are often hollow, committees that never meet, grievance processes no one uses, safeguards that protect the project, not the people.

Carbon Clarity's governance is different in four ways:

1. Power is distributed, not concentrated

Many projects have one community committee that controls everything. If that committee is captured or ineffective, there is no alternative. We distribute power across three tiers plus alternative channels, so if one pathway fails, others still work.

2. Timelines are mandatory and public

"You can file a grievance" is meaningless if responses take six months. Our 2-week response timeline is not a goal, it is a requirement. If we miss it, it is visible publicly and we must explain why and when we will respond.

3. Inclusion is structural, not aspirational

"We welcome women and youth" is nice rhetoric. Requiring minimum 3 women and 2 youth on committees, with invalid elections if quotas are not met, is structured. We do not assume inclusion will happen naturally, we design it in.

4. We designed for real constraints

We did not copy a governance model from a high-literacy, high-connectivity, stable context and drop it into Mozambique. We designed for: low literacy (visual materials), limited connectivity (SMS works when internet doesn't), fragile trust (multiple entry points, no-retaliation safeguards), and displacement dynamics (IDP/host sensitivity, mobility, insecurity).

This is governance built for the real world communities live in, not the ideal world we wish existed.

Testing Governance - What We Will Measure

How will we know if governance is working or just performing?

Metric 1: Participation rates by demographic

- Are women, youth, people with disabilities participating at rates proportional to their population?
- If not, why? (We ask directly, not assume.)

Metric 2: Grievance usage and satisfaction

- How many grievances filed per quarter? (If zero, that's a red flag—means people don't trust the system)
- How many resolved within 2 weeks?
- Of resolved grievances, what % of complainants say they were satisfied with the outcome? (Survey follow-up)

Metric 3: Committee effectiveness

- How many Benefit Fund Committee meetings held? (Should be minimum 4 per year)
- How many decisions made? (Projects approved, formula reviews, cost challenges)
- Do community members know who their committee representatives are and how to contact them? (Random survey)

Metric 4: Power in action

- Has the Benefit Fund Committee ever successfully challenged a developer cost claim or margin?
- Have VSLA Carbon Liaisons ever caught and corrected a payment error?
- Have communities ever renegotiated the benefit-sharing formula?

If the answer to Metric 4 is always "no," that suggests governance is symbolic. Real power means sometimes using it to say no, to demand changes, to hold others accountable. If committees never challenge anything, they are rubber stamps, not oversight.

SECTION 8: IMPACT & SCALE

From Transparency to Transformation

Why Impact Must Be Defined Before We Start

There is a dangerous habit in development work: launching a project, keeping busy for three years, and then scrambling at the end to figure out what impact was achieved. By then it is too late to course-correct, too late to gather baseline data, and too late to know if the changes communities experienced were because of your work or despite it.

Carbon Clarity will not make that mistake. We are defining impact now—before the pilot begins, so we know what we are aiming for, how we will measure it, and what success actually looks like. This is not just good practice. It is accountability to the communities we serve.

But here is the key question: what is the impact we are trying to create? The easy answer would be "transparent carbon payments." But that is an output, not an impact. Transparency is the tool. Impact is what transparency makes possible in people's lives.

So let us be clear about what we are actually trying to achieve.

Three Types of Impact - And Why They Matter in This Order

IMPACT 1: INFORMED AGENCY (The Foundation)

What this means:

Communities understand carbon markets well enough to make informed decisions, verify if they are being treated fairly, and take action when they are not.

This is the foundation because everything else depends on it. If communities do not understand what carbon credits are or how pricing works, then transparency just overwhelms them with meaningless numbers. But if they understand, transparency becomes power.

What this looks like in practice:

- A woman in a resettlement site can explain to her neighbor: "Carbon credits are the value created when we use less firewood. Buyers pay for that value. We should get most of it."
- When a payment arrives, she checks her receipt against the price board and knows if the numbers match or something is wrong.
- If something is wrong, she knows exactly who to tell and what will happen next.

How we measure it:

After 12 months of Carbon Clarity Hub education:

- Target: 70% of surveyed households can correctly answer 3 out of 4 basic questions:
  - What are carbon credits?
  - How long does it take from stove distribution to payment?
  - Where can you check if your payment is fair?
  - What do you do if you think your payment is wrong?

Why this matters:

Informed agency is not just about knowledge—it is about confidence. A household that knows what to expect, how to check, and how to challenge is no longer a passive beneficiary. They become an informed participant who can protect themselves. That shift from passive to active is the first impact we are trying to create.

IMPACT 2: VERIFIED FAIRNESS (The Core Promise)

What this means:

Communities receive a demonstrably fair share of carbon revenues, and they can verify that fairness themselves—not just trust someone else's word.

What this looks like in practice:

- Benefit-sharing formulas are published: "From each $10 of carbon revenue: $1.50 covers verified costs, $0.50 is developer margin (capped), $6 goes to you, $2 goes to community fund."
- Price boards show actual numbers every quarter: "This quarter: 4,000 credits sold at $12 each. Your community share: $27,600."
- Receipts show household-level breakdown: "Your 1.5 credits at $12 = $18. Costs $7.65. Your payment: $7.25. Fund: $3.10."
- Communities can check if the promised share (e.g., 57.5%) matches what was delivered.

How we measure it:

After 36 months (when payments begin):

- Target 1: Communities receive ≥50% of gross carbon revenue (vs. 10-30% typical in opaque projects)
- Target 2: 80% of surveyed households report they understand how their payment was calculated
- Target 3: Payment amounts match published formula with <5% error rate
- Target 4: Zero uncorrected discrepancies between price board promises and actual distributions

Why this matters:

This is the promise that justifies Carbon Clarity's existence. If we educate communities but they still receive unfair shares, we have failed. If they receive fair shares but cannot verify it, they still depend on trust, not transparency. We have to deliver both: fair share AND verifiable proof.

IMPACT 3: SUSTAINED TRUST AND PARTICIPATION (The Long-Term Goal)

What this means:

Because communities are informed and can verify fairness, trust in carbon markets grows over time instead of collapsing. This makes long-term participation and future scale possible.

What this looks like in practice:

- Households continue using stoves consistently (not selling them or reverting to traditional methods) because they understand and benefit from the carbon value.

- Communities actively participate in monitoring (allowing field staff visits, answering surveys) because they see the connection between data and payment.
- When new carbon opportunities appear (agroforestry, REDD+, renewable energy), communities engage willingly because previous carbon experience was positive and transparent.
- IOM can scale to additional sites because the model works and communities request it.

How we measure it:

After 36 months:

- Target 1: Stove usage rates remain ≥80% (vs. 50-60% dropout in some projects)
- Target 2: 75% of surveyed households report they "trust that carbon payments are calculated fairly" (vs. <40% in projects without transparency)
- Target 3: <10% household opt-out rate over 3 years
- Target 4: Neighboring communities request to join the model

Why this matters:

Africa's carbon potential depends on sustained community participation over decades, not one-time projects. If the first generation of carbon projects leaves communities feeling exploited or confused, the second generation will struggle to find participants. Trust is not a soft outcome, it is the foundation that makes scale possible.

Theory of Change - How Transparency Creates These Impacts

Let us map the logic explicitly, because impact does not happen automatically, it flows through a clear causal chain:

IF we provide accessible carbon education through VSLAs, visual guides, and radio (Carbon Clarity Hub)...

THEN communities will understand what carbon credits are, how pricing works, and what to expect (Informed Agency)...

AND IF we provide transparent pricing tools (price boards, itemized receipts, VSLA oversight) aligned with enforceable governance (Benefit Fund Committees, grievance mechanisms)...

THEN communities will be able to verify they are receiving fair shares and challenge when they are not (Verified Fairness)...

AND IF communities see that transparency works (issues are resolved, payments match promises, their voice matters)...

THEN trust builds, participation remains high, and communities engage in future carbon opportunities (Sustained Trust)...

WHICH ENABLES IOM to scale the model to more sites, developers to access premium buyers, and Africa's carbon markets to grow on a foundation of integrity instead of exploitation (Scale)...

RESULTING IN more households earning fair income from carbon, more forests protected, more youth employed, and more communities empowered to negotiate their participation in climate finance.

Key assumptions we are testing:

1. Education can create understanding even in low-literacy contexts (if delivered through trusted channels like VSLAs, using visual and oral methods)
2. Transparency creates accountability when paired with governance (information alone is not enough; communities need power to act on it)
3. Fair treatment builds trust over time (if communities see transparency works, they stay engaged)
4. Successful pilots can scale (if the model works in 9 sites, it can work in 50+)

If any of these assumptions prove false, we adjust the model. That is why measurement matters, not just to report impact, but to learn and adapt.

Expected Outcomes - Specific and Measurable

Here are the concrete outcomes we expect to achieve in IOM Mozambique's 2,770-household pilot over three years:

ECONOMIC OUTCOMES:

For households:

- Average carbon income per household: $50-$150 over 5-year project cycle (vs. $0-30 in failed or opaque projects)
- Households can verify their payment is fair (80%+ can explain how it was calculated)
- Payment timing is predictable (quarterly, on schedule 90%+ of the time)

For community:

- Community fund generates $20,000-$40,000 over 5 years
- Fund finances 3-5 major community projects (water, education, health) decided by elected committees
- Fund spending is transparent and audited (100% of spending documented publicly)

For developers:

- Project success rate increases (fewer failures due to community disengagement)
- Credits sell at premium prices (+$2-5 per credit) due to documented community benefits
- Developer reputation improves, enabling access to more buyers and more sites

SOCIAL OUTCOMES:

Empowerment:

- 70%+ of households understand carbon markets after 12 months of education
- Women, youth, and vulnerable groups are represented in decision-making (quotas met 100% of election cycles)
- Community members feel they have voice and agency (75%+ agree: "I can influence how carbon revenue is used")

Trust:

- 75%+ of households trust carbon payments are calculated fairly (vs. <40% baseline in opaque projects)
- Grievances are resolved quickly (90%+ within 2-week timeline)
- Communities report positive relationship with developer and IOM (70%+ satisfaction)

Inclusion:

- Women receive carbon payments directly to their mobile money accounts (100%)
- Youth participate in governance (Benefit Fund Committees include youth 100% of terms)
- People with disabilities can access information and participate (adaptive materials provided 100% of time when needed)

ENVIRONMENTAL & CLIMATE OUTCOMES:

Stove adoption and usage:

- 80%+ of distributed stoves still in use after 36 months (vs. 50-60% in some projects)
- Consistent usage enables accurate carbon accounting and higher credit issuance

Emissions reductions:

- 4,000-8,000 tCO_2e reduced over 3 years from 2,770 households
- Co-benefits documented: improved health (reduced indoor smoke), time saved (less firewood collection), forest protection

Youth employment and deforestation:

- 50-100 youth engaged in briquette/sustainable charcoal production
- 500-1,000 tonnes of briquettes produced annually, reducing wood demand

- ❖ Additional 750-1,500 tCO_2e from avoided deforestation and improved production methods

SYSTEMIC OUTCOMES:

Model validation:

- ❖ Carbon Clarity model successfully piloted and documented (full case study, lessons learned)
- ❖ IOM Mozambique can replicate in additional sites with confidence
- ❖ Other IOM country offices and partners request model adaptation

Market influence:

- ❖ Gold Standard or Verra references Carbon Clarity as example of best-practice transparency
- ❖ Carbon buyers prioritize projects with transparent benefit-sharing (market demand shifts)
- ❖ Other developers adopt similar tools to remain competitive

Policy impact:

- Mozambican government references model in national carbon market or benefit-sharing guidelines
- Regional bodies (SADC, African Union) feature as example of high-integrity community carbon engagement

What Success Does NOT Look Like

It is also important to define what we are NOT trying to achieve, because chasing the wrong metrics leads to perverse outcomes:

We are NOT trying to maximize carbon credit volume at any cost. If pushing for higher volumes means compromising transparency, rushing education, or excluding vulnerable groups, we slow down. Quality over quantity.

We are NOT trying to make communities dependent on carbon income. Carbon payments are supplemental income ($50-150 over 5 years), not livelihoods. If households are using stoves only for carbon money and not because stoves are useful, something is wrong.

We are NOT trying to eliminate all grievances. Zero grievances might mean the system works perfectly, or it might mean people are too afraid or too confused to complain. We want to see some grievances (sign of healthy accountability) resolved quickly and fairly.

We are NOT trying to make every community member a carbon expert. Basic literacy (know what credits are, how to check fairness, where to complain) is enough. Deep technical knowledge is not required for informed participation.

Success is informed, empowered communities receiving fair shares and choosing to stay engaged, not perfect participation by carbon PhDs.

Scalability - Four Pathways to Growth

Carbon Clarity is designed to scale. Here is how:

PATHWAY 1: GEOGRAPHIC EXPANSION WITHIN MOZAMBIQUE

Phase 1 (Years 1-3): Pilot with 2,770 households across 9 IOM sites

Phase 2 (Years 4-5): Scale to 15-20 additional sites, reaching 8,000-10,000 households

Phase 3 (Years 6-10): National coverage across displacement-affected areas, 25,000+ households

What enables this:

- Proven model with documented impact
- Standardized tools (training modules, visual guides, governance templates) that can be replicated
- Economies of scale (per-household cost drops from $17/year to $8-10/year)
- IOM Mozambique already operates in these areas (infrastructure exists)

Challenges to address:

- Different languages (adapt materials to Sena, Ndau, Makonde in addition to Emakhuwa)
- Varying VSLA penetration (some sites may need to strengthen VSLAs first)
- Security constraints in active conflict zones (phased entry only when safe)

PATHWAY 2: REPLICATION IN OTHER COUNTRIES

Target countries: Kenya, Uganda, Tanzania, DRC, Malawi, Ethiopia (anywhere with IOM/UNHCR operations + community carbon projects)

Model: Franchise/partnership approach

- Carbon Clarity provides: training, tool templates, quality standards, technical support
- Local partner (IOM country office or local NGO) delivers: using their staff, adapting to local language and context

- Revenue model: local partner pays licensing fee ($5,000-10,000/year) + Carbon Clarity provides technical assistance

What enables this:

- IOM operates in 100+ countries; if model works in Mozambique, other offices will request it
- UNHCR,

WFP, FAO also run cookstove and carbon projects in displacement settings

- Regional bodies (IGAD, SADC) can promote as best practice

Challenges to address:

- Each context requires adaptation (language, governance structures, carbon standards vary)
- Quality control (ensuring replication maintains transparency and safeguards)
- Capacity building (local partners need training to deliver model effectively)

PATHWAY 3: SECTORAL EXPANSION BEYOND COOKSTOVES

Target activities:

- Agroforestry (communities plant trees for carbon sequestration)
- REDD+ (communities protect forests from deforestation)
- Renewable energy (solar, biogas)
- Sustainable agriculture (soil carbon)

What stays the same:

- Education approach (VSLA-based, visual, multi-channel)
- Transparency tools (price boards, receipts, governance)
- Safeguards (FPIC, inclusion, no retaliation)

What adapts:

- Technical content (tree carbon vs. stove carbon)
- Monitoring methods (forest surveys vs. stove usage)
- Benefit-sharing formulas (may differ by activity type)

Challenges to address:

- More complex carbon accounting for some activities (REDD+ is harder than cookstoves)
- Longer timelines (trees take 5-10 years vs. stoves generating credits in 2-3 years)

PATHWAY 4: BECOMING MARKET INFRASTRUCTURE

Long-term vision: Carbon Clarity becomes the standard transparency infrastructure for community carbon projects across Africa.

What this looks like:

- Gold Standard or Verra incorporates our transparency requirements into their certification criteria: "All community carbon projects must provide itemized receipts, public price boards, and VSLA-linked oversight—or demonstrate equivalent transparency."
- Developers seeking high-integrity certification contract Carbon Clarity (or equivalent) to provide transparency services.
- Carbon buyers prioritize credits from transparent projects, creating market demand.
- Eventually, opaque projects cannot find buyers—transparency becomes market requirement, not optional

PATHWAY 4: BECOMING MARKET INFRASTRUCTURE (continued)

What enables this:

- Proven track record over 5-10 years showing transparent projects have higher success rates, better community outcomes, and fewer failures
- Growing pressure from buyers, investors, and governments for integrity and documented benefit-sharing
- Carbon Clarity documentation and tools become open-source resources that others can adapt
- Training and certification services become revenue-generating at scale

Challenges to address:

- Takes 10+ years to shift market norms
- Requires influencing standard-setters, not just implementing projects
- Risk of competitors copying surface features (price boards) without depth (governance, safeguards)

Why this matters:
If Carbon Clarity becomes market infrastructure—the default expectation, not the exception—then we have succeeded beyond our pilot. The goal is not to be the only organization providing transparency, but to make transparency so normal that every organization must provide it.

The Scaling Logic - Why This Model Can Grow

Not every pilot can scale. Some solutions work in one place but fall apart when replicated. Here is why Carbon Clarity is designed to scale:

REASON 1: WE BUILD ON EXISTING INFRASTRUCTURE

We do not create new community structures—we use VSLAs that already exist and are already trusted. We do not require new technology—we use mobile money, radio, and printed materials that already work in low-resource contexts. We do not need expensive specialists—we train local facilitators who already work with communities.

This means we can enter a new site and be operational in 3-6 months, not 3-6 years. And it means our model works anywhere VSLAs and basic mobile infrastructure exist—which is most of rural Africa.

REASON 2: OUR COSTS DECREASE AS WE SCALE

Year 1 pilot: $51 per household over 3 years.

By Year 5 at 10,000 households: ~$25-30 per household (materials are reused, training systems are standardized, staff efficiency improves).

At scale (50,000+ households): ~$15-20 per household.

This is the opposite of many interventions where per-unit costs stay high or increase. Our costs drop because we are building systems and tools once, then deploying them repeatedly.

REASON 3: ALL ACTORS BENEFIT FROM TRANSPARENCY

Scale happens when everyone involved wants more of something, not just one actor.

- Communities want it: informed participation, fair shares, voice
- Developers want it: reduced risk, premium prices, better reputation
- IOM wants it: can engage in carbon finance without compromising humanitarian principles
- Buyers want it: credible community benefits, lower reputational risk
- Governments want it: aligns with national integrity and benefit-sharing goals

When everyone benefits, scaling becomes a pull ("we want this") rather than a push ("you should adopt this"). That makes growth organic and sustainable.

REASON 4: WE DESIGNED FOR ADAPTATION, NOT REPLICATION

We are not saying "copy Mozambique exactly." We are providing a framework—education + transparency + governance—that can be adapted to local languages, governance structures, carbon activities, and cultural contexts.

Uganda might use different community structures than Mozambique. Kenya might have higher literacy and can use more text, less visuals. DRC might need stronger conflict-sensitivity protocols. The framework is flexible enough to adapt while maintaining core principles: accessibility, verifiability, enforceability.

REASON 5: IMPACT IS DEMONSTRABLE

After 3 years in Mozambique, we will have data:

- X% of households understand carbon markets
- Y% can verify fair payments
- Z% trust the system
- Community fund generated $____ and financed __ projects
- Grievances resolved in __ days on average
- Zero retaliation incidents
- Zero projects failed due to community disengagement

That evidence is what convinces the next partner to adopt the model. Scale happens through proof, not promises.

Risks to Scale - And How We Mitigate Them

Scaling is not automatic. Here are the main risks and our mitigation strategies:

RISK 1: LOSS OF QUALITY DURING REPLICATION

What could go wrong:
As we scale to more sites or partners replicate the model, quality could degrade. Price boards get posted but not updated. Grievance mechanisms exist on paper but responses take months. Quotas are met but women and youth on committees are tokens with no real voice. The model becomes performative, not transformative.

How we prevent this:

- Quality standards: Written clearly with measurable criteria (e.g., "Price boards must be updated within 2 weeks of credit sales, not quarterly").
- Certification: Partners replicating the model must meet standards to use "Carbon Clarity" name. We audit compliance.
- Community feedback loops: We survey communities directly, not just ask partners "how's it going?" If communities report transparency is not working, we investigate and support or, if necessary, revoke certification.

- Open documentation: By publishing detailed implementation guides, we reduce variance. Anyone can see what "good" looks like.

RISK 2: DEVELOPER RESISTANCE

What could go wrong:
Developers who profit from opacity resist the model. They refuse to publish prices, provide itemized receipts, or allow community oversight. If major developers refuse, the model cannot scale in carbon markets.

How we prevent this:

- IOM leverage: IOM can make transparency a partnership requirement. Developers who want access to IOM communities must provide it.
- Market pressure: Buyers increasingly demand proof of fair benefit-sharing. Transparent projects can access buyers opaque projects cannot.
- Demonstrate value to developers: After Mozambique pilot, we show developers: "Projects with Carbon Clarity had 80% stove retention vs. 55% without. That's $200,000 more in issued credits. Transparency pays."
- Start with willing partners: We don't fight resistant developers in Year 1. We work with those who see the value, prove the model, then let market dynamics pressure the resisters.

RISK 3: CONTEXT-SPECIFIC BARRIERS

What could go wrong:
Model works in Mozambique but hits barriers elsewhere: governments restrict community organizing, conflict makes fieldwork impossible, VSLAs don't exist, mobile money penetration is too low, developers are state-controlled and non-transparent by design.

How we prevent this:

- Context assessments before entry: We don't assume model works everywhere. Before replicating in a new country, we assess: Are VSLAs or equivalent structures present? Is mobile infrastructure functional? Is political environment conducive?
- Adaptive design: If VSLAs don't exist but farmer cooperatives do, we adapt. If mobile money is limited but cash is normal, we adjust payment methods.
- Strategic site selection: We prioritize countries where enabling conditions exist. We don't force the model into contexts that aren't ready.

RISK 4: DEPENDENCY AND EXIT

What could go wrong:
Communities become dependent on Carbon Clarity's presence. When we exit to scale elsewhere, transparency collapses because it was our staff, not community capacity, holding the system together.

How we prevent this:

- Capacity building from Day 1: VSLA Carbon Liaisons and Benefit Fund Committees are community members, not our staff. We train them, but they do the work.
- Handover planning: By Year 3, our role shifts from "doing" to "supporting." Local VSLA facilitators run trainings. Committees operate independently. We provide backup, not frontline delivery.
- Sustainability mechanism: By Year 3-4, a small local team (1-2 people) can maintain the model (update price boards, log grievances, support committees) funded by developer service fees. We help set this up before we exit.
- Exit criteria: We don't leave until: (1) committees meet independently, (2) VSLA Liaisons handle grievances without our support, (3) communities report they can sustain transparency mechanisms themselves.

What Scale Means for Africa's Carbon Markets

If Carbon Clarity scales successfully over the next decade, here is what changes:

FOR COMMUNITIES:

- Hundreds of thousands of households across Africa participate in carbon markets with informed consent, verified fairness, and real voice.
- The next generation of carbon projects does not repeat the exploitation patterns of the past.
- Communities have a model they can demand: "If other places get transparent pricing, we want it too."

FOR CARBON MARKETS:

- High-integrity becomes the norm, not the niche. Opaque projects struggle to find buyers.
- Africa's carbon potential ($6 billion by 2030) can be realized because communities trust the system enough to participate at scale.
- Benefit-sharing is no longer vague promises—it is verifiable, documented, enforceable.

FOR DISPLACEMENT RESPONSE:

- Carbon finance becomes a viable tool for IOM, UNHCR, and humanitarian actors—not something they avoid because it risks exploitation.
- Displaced and host communities see tangible income from climate action, not just aid.

- The model proves that vulnerable populations can participate in markets with dignity and power, if systems are designed for them.

FOR CLIMATE FINANCE GLOBALLY:

- Africa demonstrates that community carbon markets can work at scale with integrity, setting an example for Asia, Latin America, and Pacific regions.
- Standards and policies shift: transparency and verifiable benefit-sharing become requirements, not aspirations.

This is not guaranteed. Scale requires consistent execution, political will, market pressure, and community demand. But if we do this right, Carbon Clarity does not just help one project in Mozambique—it helps reset the baseline for how carbon markets engage communities across an entire continent.

SECTION 9: IMPLEMENTATION ROADMAP

From Concept to Reality - The Path Forward

Why a Roadmap Matters

Good ideas fail not because they are wrong, but because implementation is poorly planned. Teams underestimate how long things take, miss critical dependencies ("we cannot do Step 3 until Step 1 is complete"), or fail to secure key partnerships early enough. By the time they realize the problem, momentum is lost and funders have moved on.

Carbon Clarity will not fail for lack of planning. This roadmap lays out exactly what needs to happen, in what order, by when, and with whom. It identifies the critical path—the sequence of

activities that, if delayed, delays everything else. And it names the partnerships, approvals, and resources that must be secured before we can begin.

This is not a wish list. This is an operational plan.

The Critical Path - Three Dependencies We Cannot Skip

Before we dive into the detailed timeline, let us name the three things that must happen first, because everything else depends on them:

DEPENDENCY 1: FORMAL PARTNERSHIP WITH IOM MOZAMBIQUE

What this means:
A signed Memorandum of Understanding (MoU) or partnership agreement between Carbon Clarity and IOM Mozambique that defines roles, responsibilities, access to communities, data-sharing protocols, and mutual commitments.

Why this is critical:
Without IOM partnership, we have no access to the 2,770 households in the nine scale-up sites. We have no legitimacy with communities (who trust IOM, not unknown outsiders). We have no leverage with carbon developers (IOM can require transparency as a condition of partnership; we cannot). And we have no safeguarding framework to operate within.

Timeline: Must be secured in Months 0-2

Status: IOM Mozambique has confirmed interest ("the best idea from the hackathon will be refined together with IOM and piloted"). We need to convert interest into formal commitment.

DEPENDENCY 2: IDENTIFICATION OF CARBON PROJECT DEVELOPER

What this means:
IOM and Carbon Clarity agree on which carbon project developer will handle registration, MRV, and credit sales for the cookstove project, and that developer agrees to work within Carbon Clarity's transparency framework.

Why this is critical:
The developer controls the carbon process, the pricing, and the payments. If they refuse to publish prices or provide itemized receipts, our transparency model collapses. We must secure developer buy-in before we start educating communities, because we cannot promise transparency we cannot deliver.

Timeline: Must be confirmed in Months 1-3

Status: IOM's previous developer had "integrity issues" and was terminated. IOM is selecting a new developer for the scale-up phase. We need to be part of that selection process to ensure transparency requirements are included in the partnership terms.

DEPENDENCY 3: BASELINE FUNDING SECURED

What this means:
At minimum, Year 1 operating budget ($45,000) secured through combination of developer service fee commitment, grant funding, or impact investment.

Why this is critical:
We cannot hire staff, produce materials, or train VSLAs without funding. Starting without full budget risks half-finished implementation, which damages credibility and wastes the opportunity.

Timeline: Must be secured in Months 2-4

Status: IOM can facilitate connections to potential funders (climate funds, humanitarian innovation grants). Developer service fee covers ~40% of Year 1 costs, leaving ~$27,000 to secure via grants.

Once these three dependencies are met, implementation can proceed. If any of the three is missing or delayed, the entire timeline shifts. That is why we front-load them in the first four months.

Implementation Phases - 36 Months to Full Operation

Here is the complete roadmap, broken into four phases:

PHASE 0: PARTNERSHIP AND PREPARATION (Months 0-4)

Objective: Secure partnerships, finalize agreements, and prepare for field launch.

Key Activities:

Month 0-1: Post-Hackathon Engagement

- Present refined Carbon Clarity proposal to IOM Mozambique leadership

- Participate in IOM's carbon developer selection process
- Present transparency requirements to shortlisted developers
- Secure IOM endorsement letter for fundraising purposes

Month 1-2: Partnership Formalization

- Negotiate and sign MoU with IOM Mozambique defining:
    - Access to 9 pilot sites and 14 VSLAs
    - Roles (IOM: facilitation/safeguarding, Carbon Clarity: transparency/education, Developer: carbon technical)
    - Data-sharing protocols
    - Joint decision-making processes
- Incorporate Carbon Clarity (legal entity) or establish fiscal sponsorship arrangement
- Open in-country office or partnership with Mozambican NGO for local presence

Month 2-3: Developer Agreement

- Finalize transparency terms with selected carbon developer:
    - Commitment to publish quarterly price boards
    - Provide itemized receipts with all payments
    - Share cost and margin data with Benefit Fund Committees
    - Pay Carbon Clarity service fee ($18,000-20,000/year)
- Sign service agreement

Month 3-4: Funding and Staffing

- Submit grant proposals to climate and humanitarian innovation funds
- Secure Year 1 baseline funding ($45,000 target)
- Recruit staff:
    - 1 Mozambique-based Coordinator
    - 2 Field Officers (one for north cluster sites, one for south)
    - Contract with local graphic designer for visual materials
    - Contract with community radio stations for airtime

Milestone at Month 4:
✅ Signed MoU with IOM
✅ Signed service agreement with carbon developer
✅ $45,000 Year 1 funding secured
✅ Core team hired
✅ Ready to begin field activities

PHASE 1: FOUNDATION - EDUCATION AND EXPECTATION SETTING (Months 4-10)

Objective: Build basic carbon literacy in all 14 VSLAs before stoves are fully distributed and before communities start expecting payments.

Key Activities:

Month 4-5: Material Development

- Design and test visual household guide (4 pages, Emakhuwa)
- Design price board template (large format, weather-resistant)
- Develop VSLA Training Module 1: Carbon Basics (flipcharts, role-play scripts)
- Print 3,000 copies of household guide
- Produce 14 price boards (one per VSLA site)

Month 5-6: VSLA Facilitator Training

- Train 14 local VSLA facilitators (selected by IOM/communities) on:
    - What are carbon credits and why they matter
    - How to use visual guides and flipcharts
    - How to facilitate participatory sessions, not lecture
    - How to handle questions they don't know answers to ("I will find out and bring answer next meeting")
- Facilitators receive training manuals, materials, and stipends

Month 6-8: Phase 1 VSLA Training Rollout

- Each VSLA receives Training Session 1: Carbon Basics
    - What are carbon credits? (Simple definition with pictures)
    - Why do cookstoves create carbon value? (Less wood = fewer trees cut = carbon stored)
    - How long does it take? (2-3 years from stove to payment)
    - What should you expect? (Not immediate money—patience required)
- Distribute household guides to all 2,770 participating households
- Install Phase 1 price boards showing: "Stage: Project registration and stove deployment. Credits not yet issued. Expected timeline: 24-30 months to first payment."

Month 7-9: Radio Program Launch

- Record and air 8 weekly radio programs (15 minutes each) covering:
    - Episode 1: "What is carbon and why does it matter?"
    - Episode 2: "Your stove creates value—here's how"
    - Episode 3: "The long journey from stove to payment"
    - Episode 4: "Why monitoring visits matter"
    - Episode 5-8: Q&A episodes answering common questions collected from VSLAs
- Programs air on community radio in Emakhuwa during high-listenership times (evening)

Month 8-10: VSLA Carbon Liaison Elections

- Each of 14 VSLAs elects one Carbon Liaison
- Carbon Liaisons receive additional training:
    - How to read and interpret price boards
    - How to collect questions from VSLA members
    - How to log and raise grievances
    - Role-play: "A member says their payment seems low—what do you do?"
- Liaisons receive contact list (Carbon Clarity phone/WhatsApp, IOM safeguarding focal point)

Milestone at Month 10:
✅ 2,770 households have received household guides
✅ 14 VSLAs have completed Training Session 1
✅ 14 Carbon Liaisons elected and trained
✅ 8 radio episodes aired
✅ 14 price boards installed and showing current project stage
✅ Baseline survey completed: 70%+ can explain "what carbon credits are" in simple terms

PHASE 2: DEEPENING UNDERSTANDING (Months 10-24)

Objective: Build deeper carbon literacy and establish governance structures while the carbon project progresses through MRV and registration.

Key Activities:

Month 10-12: Training Module 2 Rollout

- Each VSLA receives Training Session 2: Understanding Costs and Margins
    - What are legitimate costs? (Transport, verification, registry)
    - What is a margin? (What developer keeps for their work)
    - What is fair? (International standards say 50%+ to communities)
    - Activity: Practice reading sample receipts, comparing to price boards
- Expanded visual guides distributed (8 pages, adding detail on costs and formulas)

Month 12-15: Benefit Fund Committee Formation

- Community meetings in each of 9 sites to:
    - Explain role and power of Benefit Fund Committee
    - Ensure understanding of representation requirements (min 3 women, min 2 youth, etc.)
    - Conduct elections
- 9 Benefit Fund Committees elected (5-7 members each = 45-63 total committee members across sites)

- Committee members receive intensive training (2-day workshop):
    - Financial oversight basics
    - How to review quarterly reports
    - How to question costs and demand receipts
    - Governance and decision-making processes
    - How to facilitate community meetings

Month 15-18: Ongoing Radio and Updates

- Continue bi-weekly radio programs (now 26 episodes total) covering:
    - What is MRV and why does monitoring take time?
    - How third-party verification works
    - What happens during project registration
    - Stories from other carbon projects (success and failure examples)
- Price boards updated quarterly with project status:
    - "Stage: MRV data collection ongoing. 2,500 households monitored. Verification expected Month 24."

Month 18-21: Transparency Tools Testing

- Develop and test receipt templates with sample data
- Test price board updating process (how fast can we update after developer provides data?)
- Pilot grievance logging system with VSLA Carbon Liaisons
- Refine based on feedback: "Are receipts understandable? Is price board text too small? Is grievance process clear?"

Month 21-24: Training Module 3 Rollout

- Each VSLA receives Training Session 3: Your Rights and How to Use Them
    - You have right to see the numbers
    - You have right to ask questions
    - You have right to complain if something seems wrong
    - Role-play: "Your payment seems low—walk through the full process of checking and challenging"
- Carbon Liaisons practice grievance logging with real (but hypothetical) scenarios

Milestone at Month 24:
✅ All VSLAs completed 3 training sessions
✅ 9 Benefit Fund Committees elected and trained
✅ 26 radio episodes aired
✅ Price boards showing project at verification stage
✅ Transparency tools tested and refined

✅ Grievance mechanism operational (even though no payments yet)
✅ Follow-up survey: 80%+ can explain how to check if payment is fair

PHASE 3: FULL TRANSPARENCY ACTIVATION (Months 24-36)

Objective: Credits are issued and sold; full transparency infrastructure activates; communities verify fairness.

Key Activities:

Month 24-27: Credit Issuance and Price Board Activation

- Developer receives verification statement and credits are issued by registry
- Carbon Clarity receives data from developer:
    - Total credits issued
    - Sale price negotiated with buyer
    - Breakdown of costs
    - Developer margin
- Price boards updated with REAL NUMBERS for first time:
    - "4,000 credits issued and sold at $12 each. Gross revenue: $48,000. Cost breakdown: Transport $7,200, Verification $6,000, Registry $2,400, Developer margin $4,800. Community share: $27,600 (57.5%)."
- Community meetings in all 9 sites to explain the numbers publicly and answer questions

Month 27-28: First Payment and Receipt Distribution

- Developer makes first payment to households via mobile money
- Every household receives itemized receipt (paper copy + SMS) showing:
    - Their credits
    - Price per credit
    - Costs and margin
    - Their payment amount
    - Community fund deposit amount
- VSLA Carbon Liaisons check receipts of 5-10 members per VSLA, compare to price board, verify numbers match
- Any discrepancies logged immediately as grievances

Month 28-30: Grievance Resolution and Committee Oversight

- Process all grievances from first payment cycle
    - Target: 90%+ resolved within 2 weeks
    - Outcomes posted publicly on price boards
- Benefit Fund Committees meet to:

    - Review total community fund deposit
    - Verify it matches promised formula
    - Begin proposing community fund projects (water, education, health)
- First community fund project approved and spending begins (publicly documented)

Month 30-33: Second Payment Cycle

- Developer sells additional credits (ongoing quarterly or biannual cycles)
- Price boards updated with Q2 numbers
- Receipts distributed again
- Grievances processed
- Benefit Fund Committees review Q2 deposits and approve additional projects

Month 33-36: Evaluation and Documentation

- Conduct endline survey:
    - Do households understand how payments are calculated? (Target: 80%+)
    - Do they trust payments are fair? (Target: 75%+)
    - Can they explain how to file a grievance? (Target: 85%+)
- Independent audit of community fund spending
- Document case studies:
    - Amina's grievance: how it was filed, investigated, and resolved
    - Youth inclusion: how quotas worked in practice
    - Community fund project: water point funded and built, impact on daily life
- Prepare full pilot evaluation report for IOM, funders, and replication partners

Milestone at Month 36:
✅ Two full payment cycles completed with full transparency
✅ 90%+ of grievances resolved within 2-week timeline
✅ Community fund has financed 3-5 community projects
✅ 80%+ of households report understanding and trusting the system
✅ Zero retaliation incidents documented
✅ Pilot evaluation completed with documented impact
✅ IOM approves model for scale-up to additional sites

PHASE 4: HANDOVER AND SCALE PREPARATION (Months 36-48)

Objective: Transition to local ownership and prepare for geographic/sectoral scale.

Key Activities:

Month 36-40: Capacity Handover

- Train local Mozambican NGO or IOM staff to take over Carbon Clarity operations:
    - Price board updating
    - Grievance logging and follow-up
    - Committee support
    - VSLA facilitator supervision
- Carbon Clarity role shifts from implementation to technical assistance
- Local team takes over day-to-day operations with Carbon Clarity providing backup

Month 40-44: Replication Preparation

- Document full implementation

PHASE 4: HANDOVER AND SCALE PREPARATION (Months 36-48) - continued

Month 40-44: Replication Preparation

- Document full implementation guide:
    - Step-by-step manual for replicating Carbon Clarity in new sites
    - All tool templates (training modules, visual guides, price boards, receipts, governance documents)
    - Lessons learned and adaptation guidance
    - Budget templates and cost projections
- Produce case study video featuring community members explaining how transparency works in their own words
- Present findings at regional forums (SADC, African Carbon Markets Initiative convenings)

Month 44-48: Scale-Up Planning

- IOM Mozambique identifies next 10-15 sites for expansion
- Carbon Clarity secures scale-up funding (now easier with proven track record)
- Begin Phase 1 activities in new sites while maintaining support for original 9 sites
- Explore replication in other countries (Kenya, Uganda inquiries)

Milestone at Month 48:
✅ Local team independently operating in original 9 sites
✅ Carbon Clarity providing technical assistance, not direct implementation
✅ Full implementation guide and tools published
✅ Scale-up to 10-15 new sites begun
✅ At least 2 other countries/partners expressing interest in replication

Critical Milestones Summary

Here are the non-negotiable checkpoints that determine if we stay on track:

Month	Milestone	If This Fails
2	IOM partnership MoU signed	Cannot proceed—no community access
3	Developer agreement finalized	Cannot promise transparency without developer buy-in
4	Year 1 funding secured	Cannot hire staff or produce materials
10	Phase 1 education complete + Liaisons elected	Communities unprepared for transparency phase
24	Governance structures operational	Cannot enforce transparency when payments start
28	First payment with full transparency	Core model validation fails
36	Pilot evaluation shows positive impact	Cannot justify scale-up or replication

Each milestone has a go/no-go decision point: If we miss it by >1 month, we pause, diagnose why, and adjust before proceeding. Better to delay than to implement poorly.

Key Partnerships and Their Roles

Carbon Clarity cannot succeed alone. Here are the essential partners and what we need from each:

IOM MOZAMBIQUE (Lead implementing partner)

Their role:

- Provide access to 9 pilot sites and 14 VSLAs
- Co-select carbon developer with transparency requirements
- Facilitate community introductions and trust-building
- Provide safeguarding oversight and escalation pathway
- Co-fund field activities where possible

What we need from them:

- Formal partnership commitment (MoU)
- Staff time for coordination meetings (monthly)
- Integration into existing community engagement schedule
- Endorsement for fundraising and replication

Status: Interest confirmed, MoU negotiation is Month 0-2 priority

CARBON PROJECT DEVELOPER (Technical carbon partner)

Their role:

- Register project with Gold Standard or Verra
- Implement MRV system and collect stove usage data
- Hire third-party verifier
- Sell credits to buyers
- Make payments to communities
- Provide data to Carbon Clarity for price boards and receipts

What we need from them:

- Commitment to transparency terms (publish prices, provide receipts)
- Quarterly data sharing (revenues, costs, margins)
- Payment to Carbon Clarity service fee ($18-20k/year)
- Participation in Benefit Fund Committee meetings when requested

Status: To be identified in Months 1-3; IOM selecting developer for scale-up phase

VSLA FACILITATORS & NETWORKS (Community delivery channel)

Their role:

- Deliver VSLA training sessions on carbon literacy
- Support Carbon Liaison elections and ongoing liaison work
- Monitor for exclusion or tensions
- Report to Carbon Clarity and IOM on community dynamics

What we need from them:

- Time commitment (3 sessions per VSLA over 3 years + ongoing support)
- Training participation (2-day facilitator training)
- Local language fluency (Emakhuwa)
- Trusted standing in communities

Status: IOM can identify facilitators; compensation comes from Carbon Clarity budget

CARBON STANDARDS BODIES (Gold Standard, Verra)

Their role:

- Provide certification framework for cookstove carbon project
- Review and potentially adopt Carbon Clarity transparency requirements as best practice

What we need from them:

- Willingness to feature Carbon Clarity as example in guidance documents
- Consideration of incorporating our transparency tools into future certification criteria
- Technical input on aligning our tools with their safeguard requirements

Status: Engagement begins Month 12-18 once pilot is underway

GRANT FUNDERS (Climate funds, humanitarian innovation funds)

Their role:

- Provide baseline funding for education component (~$20-30k/year Years 1-3)

What we need from them:

- Flexible funding (not restricted to narrow outputs)
- Multi-year commitment (ideally 3 years to cover pilot)
- Willingness to fund "infrastructure" (education, governance) not just direct services

Potential funders: GCF Readiness, IKEA Foundation, UNHCR Innovation Fund, Open Society, climate-focused family foundations

Status: Proposals submitted Month 2-3 after IOM endorsement secured

MOZAMBICAN GOVERNMENT (Policy and enabling environment)

Their role:

- Provide policy framework for carbon projects in displacement areas
- Facilitate approvals and data collection
- Potentially adopt Carbon Clarity principles in national carbon market guidance

What we need from them:

- No active resistance to community-governed carbon models
- Access to resettlement sites through IOM channels
- Consideration of Carbon Clarity as input to national benefit-sharing policy

Status: IOM manages government relationships; Carbon Clarity supports with documentation

Resource Requirements - What This Takes

HUMAN RESOURCES:

Year 1:

- 1 Mozambique Coordinator (full-time): $20,000
- 2 Field Officers (full-time): $24,000 ($12k each)
- 14 VSLA Facilitators (part-time, contract): $8,000 stipends
- 1 M&E/Documentation specialist (part-time): $6,000
- Carbon Clarity HQ support (10% FTE): $5,000

Years 2-3: Similar structure, slight salary increases

MATERIAL RESOURCES:

- Visual guides: $5,000 (3,000 copies × 2 versions)
- Price boards: $4,000 (14 boards + installation)
- Training materials: $3,000 (flipcharts, role-play kits)
- Radio production + airtime: $8,000/year
- Receipt printing: $4,000 over Years 2-3
- Office equipment + connectivity: $4,000
- Transport (motorcycles/fuel): $6,000/year

TOTAL 3-YEAR BUDGET: ~$141,000

Cost per household: $51 over 3 years (~$17/year)

Funding sources:

- 40-50% from developer service fees
- 30-40% from grants
- 10-20% from results-based finance

Risk Mitigation - What Could Go Wrong and Our Backup Plans

RISK 1: DELAYED DEVELOPER SELECTION

Impact: If IOM takes 6-9 months to select developer instead of 3, our timeline shifts.

Mitigation:

- Begin Phase 1 education even without developer finalized (foundational carbon literacy doesn't depend on specific developer)
- Build flexibility into funding timeline (don't spend all Year 1 budget in first 6 months)
- Use delay time to strengthen VSLA relationships and refine materials

RISK 2: DEVELOPER REFUSES TRANSPARENCY TERMS

Impact: If selected developer will not publish prices or provide receipts, core model fails.

Mitigation:

- Make transparency requirements clear in IOM's developer selection criteria upfront
- Carbon Clarity participates in developer interviews/selection to assess willingness
- If developer refuses post-selection, IOM has grounds to terminate for non-compliance with partnership terms
- Worst case: pilot delayed while IOM finds alternative developer who will comply

RISK 3: FUNDING GAPS

Impact: If we secure only 50-60% of Year 1 budget, we must scale back activities.

Mitigation:

- Prioritize core activities: training and price boards over nice-to-haves like expanded radio
- Delay scale-up planning until pilot is proven (focus funding on getting 9 sites right)
- Seek in-kind support (IOM office space, transport sharing, volunteer facilitators)
- Launch crowdfunding for specific items (e.g., "Fund 3,000 household guides - $5,000")

RISK 4: COMMUNITY RESISTANCE OR LOW ENGAGEMENT

Impact: If VSLAs don't elect liaisons or households don't attend trainings, model cannot function.

Mitigation:

- Co-design with communities from Day 1 (don't impose structure from outside)
- Use trusted channels (IOM, existing VSLA facilitators) not external staff
- Listen to feedback: if trainings don't work, adapt format/timing/content
- Small incentives (Carbon Liaisons receive $5-10/quarter stipend from community fund for their time)

RISK 5: POLITICAL OR SECURITY DISRUPTION

Impact: If conflict flares in pilot sites, field activities pause.

Mitigation:

- IOM continuously monitors security; we defer to their risk assessments
- If one site becomes inaccessible, focus resources on remaining 8
- Remote support where possible (phone check-ins with Carbon Liaisons, radio continues even if field visits stop)
- Design for resumption: when access returns, we pick up where we left off

Success Indicators - How We Know We're On Track

Beyond the major milestones, here are the monthly/quarterly indicators we track:

PROCESS INDICATORS (Are we doing what we planned?):

- Training sessions delivered on schedule (target: 100% on time)
- Price boards updated within 2 weeks of new data (target: 100%)
- Grievances logged and responded to within timeline (target: 90%+ within 2 weeks)
- Committee meetings held quarterly (target: 4 per year minimum)

PARTICIPATION INDICATORS (Are people engaging?):

- VSLA training attendance (target: 70%+ of members at each session)
- Carbon Liaison election participation (target: all 14 VSLAs elect liaisons)
- Committee election turnout (target: 60%+ community members participate in elections)
- Radio listenership (tracked via sample surveys; target: 50%+ households listen to at least 1 episode)

UNDERSTANDING INDICATORS (Are people learning?):

- % who can explain "what carbon credits are" (Month 10 target: 70%+)
- % who can explain "how to check if payment is fair" (Month 24 target: 80%+)
- % who know "how to file grievance" (Month 24 target: 85%+)

TRUST INDICATORS (Are people confident in the system?):

- % who "trust carbon payments are calculated fairly" (Month 36 target: 75%+)
- % who "feel they have voice in how carbon money is used" (Month 36 target: 70%+)
- Opt-out rate (Month 36 target: <10%)

If we are significantly missing indicators (e.g., only 40% understand after training), we stop and diagnose: Is content too complex? Are facilitators not explaining well? Are language barriers bigger than we thought? Then we adapt before proceeding.

SECTION 10: LITERATURE REVIEW & EVIDENCE BASE

Standing on the Shoulders of Evidence

Why This Review Matters

Carbon Clarity is not built on hunches or good intentions. Every design choice—from using VSLAs as delivery channels to requiring itemized receipts to mandating youth quotas on committees—is grounded in evidence from research, documented practice, and field experience. This literature review shows where that evidence came from and how it shaped our solution.

We have already cited specific sources throughout this report. This section provides the synthesis: what did the literature tell us, where are the gaps, and how does Carbon Clarity contribute to filling them?

Four Bodies of Evidence That Inform Our Design

THEME 1: TRANSPARENCY GAPS IN CARBON BENEFIT-SHARING

Key sources:

- "Assessing transparency and integrity of benefit-sharing arrangements in voluntary carbon markets" (independent assessment documenting opacity)
- Forest Carbon Partnership Facility (FCPF) benefit-sharing guidance
- Plan Vivo benefit-sharing standards
- Gold Standard factsheet on inclusivity, safeguards and benefit sharing

What the literature shows:
Fewer than 30% of carbon projects publish clear benefit-sharing formulas. Even projects that promise community benefits rarely disclose: actual sale prices, itemized costs, developer

margins, or verification that promised shares were delivered. This opacity leaves communities unable to verify fairness and creates conditions for exploitation.

How this informed our design:

- Public price boards showing all numbers (price, costs, margin, community share)
- Itemized receipts with every payment
- Benefit Fund Committees with power to request documentation and challenge costs
- Quarterly transparent reporting, not annual or never

Gap we address: Most guidance says "projects should share benefits transparently" but does not specify *how* to make information accessible to low-literacy communities. We provide the operational tools (visual boards, simple receipts, VSLA-based explanation) that turn policy into practice.

THEME 2: HIGH-INTEGRITY CARBON MARKETS IN AFRICA

Key sources:

- Africa Carbon Markets Initiative (ACMI) reports
- Integrity Council for the Voluntary Carbon Market (ICVCM) Core Carbon Principles
- UNDP High-Integrity Carbon Markets Toolkit
- Ecosystem Marketplace voluntary carbon market data

What the literature shows:
Africa could generate $6 billion annually from carbon markets by 2030, but only if projects demonstrate high integrity: transparent benefit-sharing, meaningful community participation, robust safeguards, and alignment with national strategies. Current lack of trust and documented cases of exploitation risk undermining scale before it begins.

How this informed our design:

- Alignment with Gold Standard and ICVCM principles on benefit-sharing and participation
- Focus on building trust through verifiable transparency (not just promises)
- Partnership model that includes government and works within national frameworks
- Designed for scale from Day 1 (replicable tools, documented model)

Gap we address: High-integrity frameworks tell *what* should happen (fair benefit-sharing, community voice) but lack practical models for *how* to implement this in fragile, low-resource, displacement contexts. Carbon Clarity is an operational blueprint, not another policy document.

THEME 3: DISPLACEMENT CONTEXTS AND LIVELIHOODS IN MOZAMBIQUE

Key sources:

- IOM Mozambique displacement reports and energy project briefs
- World Bank energy access and literacy data for Mozambique
- UNHCR and humanitarian guidance on livelihoods in displacement settings
- CARE and Oxfam VSLA programme evaluations

What the literature shows:
Northern Mozambique has over 1 million displaced people, 95% reliance on biomass for cooking, and female literacy rates of 30-35% in affected areas. VSLAs are active and trusted in these contexts, serving 2-3 million people nationally. But displacement contexts face specific risks: mobility, insecurity, weak formal institutions, and compounded vulnerabilities that make standard development approaches fail.

How this informed our design:

- VSLA-based model leverages existing trusted structures rather than creating new ones
- Visual, oral, multi-channel education designed for low-literacy contexts
- Mobile money and local office model works with limited connectivity
- Conflict-sensitive safeguards address IDP/host tensions
- Phased rollout prevents overwhelming communities with information

Gap we address: Most carbon market literature assumes stable, literate populations with functioning institutions. We provide a model specifically designed for displacement contexts where those conditions do not exist.

THEME 4: COMMUNITY PARTICIPATION, SAFEGUARDS AND FPIC

Key sources:

- Free, Prior and Informed Consent (FPIC) frameworks for carbon projects
- Gold Standard safeguard requirements
- WHO data on indoor air pollution and gender impacts
- Research on elite capture and exclusion in community programs

What the literature shows:
Meaningful participation requires more than consultation—it requires accessible information, structured power-sharing, protection from retaliation, and specific measures to include women, youth, and marginalized groups. Without these, "community participation" becomes performative, and elite capture or exclusion undermines stated goals.

How this informed our design:

- FPIC operationalized through visual contracts, oral explanation, ongoing consent
- Mandatory quotas for women, youth, and vulnerable groups on committees
- Multiple participation channels (VSLA, SMS, anonymous grievances) so intimidated individuals have options

- No-retaliation safeguards with public monitoring
- Conflict-sensitive design to prevent harm

Gap we address: FPIC and safeguard guidance is often abstract. We show exactly what informed consent looks like in a low-literacy displacement context: one-page visual contracts, repeated oral explanations over years, election quotas, and grievance mechanisms with 2-week response times.

How We Validate Claims

Throughout this report, we have made specific claims. Here is the evidence for each:

Claim	**Evidence Source**
<30% of projects publish benefit formulas	Benefit-sharing transparency assessment
Communities often receive 10-30% of revenue	Literature review + IOM interview
Female literacy in Cabo Delgado is 30-35%	UNESCO/World Bank Mozambique education data
IOM's pilot failed due to developer integrity issues	Direct interview with IOM Mozambique (Mercy Mutavi)
VSLAs have strong internal accountability	CARE/Oxfam VSLA evaluations
14 active VSLAs in IOM project sites	IOM Mozambique confirmation
Carbon projects take 2-3 years to payment	Gold Standard/Verra project cycle documentation
Africa could generate $6B by 2030	ACMI projections
Displacement populations face 60%+ youth unemployment	IOM/UNHCR displacement and livelihoods reports
Northern Mozambique loses 0.8-1.2% forests annually	World Bank/FAO forest data

Every number, every claim, every design principle has a source. We are not guessing.

What the Literature Does NOT Tell Us (The Gaps)

The literature helped us design Carbon Clarity, but it also revealed what is missing—the gaps our work aims to fill:

GAP 1: NO OPERATIONAL MODELS FOR TRANSPARENCY IN LOW-LITERACY CONTEXTS

Guidance says "provide transparent information" but does not show *how* when 70% of communities cannot read contracts. We provide the first documented model using visual boards, oral explanation, and VSLA-based verification.

GAP 2: LIMITED RESEARCH ON CARBON MARKETS IN DISPLACEMENT SETTINGS

Most carbon literature focuses on indigenous communities or stable rural populations. Displacement contexts—with mobility, trauma, weak institutions, and IDP/host tensions—are under-researched. We fill this gap with a model designed specifically for resettlement sites.

GAP 3: WEAK LINK BETWEEN POLICY AND PRACTICE ON BENEFIT-SHARING

Standards require benefit-sharing plans, but audits rarely verify if communities actually received what was promised or understood the arrangement. We provide tools that make verification possible at the community level, not just the auditor level.

GAP 4: NO DOCUMENTED MODELS USING VSLAs FOR CARBON GOVERNANCE

VSLAs are well-researched for savings and loans, but rarely for environmental governance. We show how VSLA structures can extend to carbon oversight, leveraging existing accountability culture.

Carbon Clarity's contribution to the literature will be: a fully documented, field-tested model showing *how* to implement transparent, community-governed carbon markets in displacement contexts. If we succeed, the next project does not have to start from scratch—they can adapt our blueprint.

Methodology Note

Our literature review combined:

- Academic and grey literature searches (Google Scholar, organizational reports)

- Standards documentation (Gold Standard, Verra, ICVCM publicly available guidance)
- Direct field data (IOM Mozambique interview, project briefs)
- Policy and advocacy reports (ACMI, UNDP, climate funds)

We prioritized sources that were: recent (2020-2025), context-relevant (Africa, displacement, community carbon), and methodologically sound (peer-reviewed or from credible institutions). Where evidence was limited (e.g., carbon markets in IDP contexts), we relied on closest analogues (carbon markets in vulnerable communities; livelihoods programming in displacement) and adapted with caution.

SECTION 11: CONCLUSION

From Transparency to Trust - Why Carbon Clarity Matters

What We Set Out to Do

We began this work with a simple observation: communities in northern Mozambique are being asked to participate in carbon markets they cannot see, understand, or negotiate. They use improved cookstoves every day. They generate carbon value through their actions. But when—or if—carbon money flows, it arrives as a mystery. No breakdown. No explanation. No way to verify fairness.

And when carbon projects fail due to developer misconduct, as happened in IOM's pilot, communities are left with nothing and no clear understanding of what went wrong.

This is not how markets are supposed to work. Markets require information symmetry, transparency, and the ability to negotiate. When one party controls all the information and the

other party is told to trust blindly, that is not a market—it is exploitation dressed in market language.

Carbon Clarity exists to close that gap. Not by replacing carbon markets, but by making them work the way they are supposed to: with informed participants, transparent pricing, and enforceable accountability.

What We Built

Over the course of this hackathon, we designed a transparency and education infrastructure specifically for displacement contexts. It has two integrated components:

The Carbon Clarity Hub teaches communities what carbon credits are, how the process works from registration to payment, what fair benefit-sharing looks like, and what their rights are. It uses trusted channels—VSLAs, community radio, visual guides—and delivers information in accessible formats designed for low-literacy, mixed-language contexts.

The Transparency Tools make the carbon process visible and checkable: public price boards showing prices, costs, and margins; itemized receipts with every payment; VSLA Carbon Liaisons who review numbers on behalf of communities; Benefit Fund Committees with power to question costs and renegotiate formulas; and clear grievance mechanisms with 2-week response guarantees.

Together, these components transform communities from passive beneficiaries into informed participants who can protect themselves.

We grounded this design in the real context of IOM Mozambique's cookstove scale-up project: 2,770 households, nine sites, 14 active VSLAs, 2-3 year timeline from stove distribution to carbon payment. We learned from IOM's failed pilot. We aligned with Gold Standard and Verra frameworks rather than trying to replace them. We designed for the actual constraints communities face: 30-35% female literacy, limited connectivity, displacement-related mobility and insecurity, and fragile trust in outsiders.

And we built a complete system: not just tools, but governance structures, safeguards, business model, implementation roadmap, risk mitigation strategies, and measurable impact indicators.

This is not a concept. This is a blueprint ready to pilot.

Why This Solution Can Win

Hackathons produce many good ideas. Here is why Carbon Clarity deserves to advance and be implemented:

FIRST: WE SOLVE A PROVEN, URGENT PROBLEM

IOM Mozambique experienced this problem directly. Their pilot failed. Communities lost out. And as IOM scales to 2,770 new cookstoves, they explicitly want to avoid repeating that failure. We are not solving a hypothetical problem—we are solving the problem IOM lived through and told us they need help addressing.

SECOND: WE HAVE A CONFIRMED IMPLEMENTATION PARTNER

IOM Mozambique stated: *"The best idea from the hackathon will be refined together with IOM and our implementing partners and piloted during the project scale-up with a small group of communities."* We have a real partner, a real timeline (scale-up happening in 2026), and real communities ready to pilot. This is not "if we find a partner someday"—the partner exists and is waiting.

THIRD: WE WORK WITHIN THE SYSTEM, NOT AGAINST IT

We do not ask Gold Standard to change their rules or IOM to abandon their processes. We provide tools that make existing requirements—fair benefit-sharing, community participation, safeguards—actually work in practice. This makes adoption easier because we are helping partners do what they already want to do, not asking them to do something completely new.

FOURTH: WE DESIGNED FOR REAL CONSTRAINTS

Low literacy. Multiple languages. Limited connectivity. Displacement mobility. Fragile trust. Elite capture risks. Information overload. We did not ignore these challenges—we designed around every single one. Visual materials for literacy. Phased rollout for overload. VSLA channels for trust. Quotas and alternative channels for inclusion. This is not an ideal-world solution dropped into a hard context—it is built for the context from the ground up.

FIFTH: WE HAVE A VIABLE BUSINESS MODEL

We showed how Carbon Clarity sustains itself financially: developer service fees (40-50%), grants for education (30-40%), results-based finance (10-20%). Communities pay nothing. The model moves from 60% grant-funded in Year 1 to 70% service-fee-funded by Year 5. This is a path to long-term sustainability, not permanent dependency.

SIXTH: WE CAN DEMONSTRATE IMPACT

By Month 36, we will have measurable outcomes: X% carbon literacy, Y% verified fairness, Z% community trust, $ community fund projects delivered, grievances resolved in <2 weeks. This is not "we hope to make communities happier"—we have concrete metrics judges and funders can verify.

SEVENTH: WE ARE DESIGNED TO SCALE

From 9 sites in Mozambique to 50+ sites across Africa. From cookstoves to agroforestry, REDD+, renewable energy. From pilot to market infrastructure. Every phase of our roadmap includes the next level of scale. This is not a one-off project—it is a model built to grow.

What Success Looks Like Five Years From Now

If Carbon Clarity succeeds, here is what we will see in 2031:

In Mozambique:

- 10,000+ households across 20+ sites participating in transparent carbon projects
- Community funds have generated $200,000+ financing water, education, and health projects communities chose
- Zero carbon project failures due to developer integrity issues in Carbon Clarity sites
- IOM Mozambique using our model as standard practice for all carbon-linked programming

Across Africa:

- Carbon Clarity operating in Kenya, Uganda, Tanzania, DRC, Malawi, Ethiopia (6+ countries)
- 50,000+ households benefiting from transparent carbon finance
- Regional carbon market initiatives (ACMI, SADC) referencing our model as best practice

In carbon markets:

- Gold Standard or Verra incorporating our transparency tools into certification requirements
- Carbon buyers prioritizing credits from transparent projects
- "Transparent benefit-sharing" shifting from niche to norm

For communities:

- Women, youth, and vulnerable households earning fair income from carbon
- Communities demanding transparency in all climate finance, not just carbon
- Next generation of carbon projects designed with Carbon Clarity principles from Day 1

That is the world we are building: where transparency is infrastructure, not innovation.

The Choice Ahead

Africa's carbon markets are at a crossroads. One path leads to rapid scale built on opacity, extraction, and broken trust, high volumes in the short term, collapse in the long term. The other

path leads to slower, deliberate growth built on transparency, fairness, and sustained community trust, stable participation that can last decades.

Carbon Clarity is the infrastructure for the second path. We cannot force the market to choose transparency. But we can prove it works. We can show that communities who understand carbon markets participate more consistently, that transparent projects have higher success rates, and that fairness and financial viability are compatible.

IOM Mozambique is ready to test this. They are distributing 2,770 cookstoves in 2026. They want transparency infrastructure to prevent another failed pilot. They have 14 VSLAs ready to engage.

The opportunity is here. The partner is confirmed. The timeline is set.

The question is not whether Carbon Clarity is needed. IOM's failed pilot proves it is. The question is whether we build it now—before the next 2,770 households sign contracts they cannot read, before the next carbon project fails due to integrity issues, before Africa's carbon potential is undermined by the same opacity that has broken trust in markets for generations.

Our Ask

We are asking for three things:

1. Selection for the final Demo Day

Advance Carbon Clarity to the final round so we can develop this solution in partnership with IOM Mozambique and present a fully detailed implementation plan to stakeholders and potential funders.

2. Connection to key stakeholders

Facilitate introductions to:

- IOM Mozambique leadership and the carbon developer they are selecting
- Potential funders (climate funds, humanitarian innovation funds, impact investors)
- Gold Standard or Verra staff who can advise on aligning our tools with certification requirements
- Other displacement-focused organizations interested in replicating the model

3. Support to pilot in 2026

Help us move from concept to implementation during IOM's cookstove scale-up phase. This means:

- Partnership formalization by March 2026

- Baseline funding secured by April 2026
- Field activities beginning by May 2026
- First VSLA trainings completed by August 2026

A Final Thought

Throughout this work, we kept coming back to one person: Francisca. A displaced mother in Cabo Delgado. Cooking with a stove that creates carbon value. Told she will benefit. But given no information, no receipts, no way to verify if the promise is kept.

In some ways, Francisca is just a persona—a composite we built from project documents and research. But in the ways that matter, she is very real. There are thousands of women exactly like her in northern Mozambique right now, using stoves, creating climate value, and waiting to see if this time—this carbon project, this developer, this promise—will be different from all the others that left them with nothing.

Carbon Clarity is our answer to Francisca. It says: you deserve to understand. You deserve to verify. You deserve power over decisions that affect your household. You deserve more than trust—you deserve proof.

If we get this right, Francisca will not need to trust blindly. She will check her receipt against the price board, see the numbers match, and know she was treated fairly. She will tell her neighbor how the system works. She will vote for the next Benefit Fund Committee representative. She will file a grievance if something feels wrong—and see it resolved in two weeks.

That shift—from powerless to informed, from passive to active, from exploited to protected—is what Carbon Clarity makes possible.

And if it works for Francisca in Mozambique, it can work for millions of others across Africa.

That is the carbon market we are trying to build. One receipt, one price board, one VSLA meeting, one verified payment at a time.

12. APPENDICES
- Visual contract sample
- Receipt template
- Interview notes summary
- References

TOTAL: ~20-25 pages